FINSIGHT AI / ENGINEERING REFERENCE

Database Schema and ERD Reference

A complete reference for the core PostgreSQL schema, optional trading extension, table columns, primary keys, foreign keys, and ER models.

158 Core tables	15 Optional trading tables	173 Total target tables
Research, market data, RAG, portfolios, DCF, risk, workflows and backtesting.	Paper trading and broker connectivity; do not apply in Release 1.	Full target-state reference, not a requirement to ship everything at once.

What is inside

1. Database architecture and implementation boundary
2. All available ER models and the cross-domain dependency map
3. Domain-by-domain table schemas, including columns, data types, PKs and FKs
4. Core vs optional schema files and validation notes

Source basis: FinSight_Software Requirements Specification (SRS) and the schema/catalogue files generated for the FinSight DB Design package. DCF and quantitative-risk detail reflects the additional scope supplied for this design.

1. How to use this reference

This document describes the schema as declared in the supplied SQL files. The table inventory is generated from the CREATE TABLE definitions, and the domain descriptions and design notes are drawn from the accompanying table catalogue and README.

Source files

File	Purpose
`schema.sql`	Core PostgreSQL target schema: 158 tables for identity, providers, market/reference data, RAG, portfolios, DCF, risk, workflows and backtesting.
`schema_optional_trading.sql`	Optional Phase 2+ extension: 15 tables for paper trading, broker connections, orders, fills, risk checks, approvals and reconciliation.
`seed_reference_data.sql`	Development seed rows for selected lookup/reference tables; it is not a production data feed.
`table_catalog.md`	Purpose descriptions, primary-key and foreign-key inventory, implementation boundaries and SRS traceability.
`erd/*.svg / *.mmd`	Vector diagrams and editable Mermaid ERD sources; see the ERD section below.

Key interpretation rules

PostgreSQL is the structured source of truth

Document ownership, permissions, instrument identifiers, price/fundamental provenance, valuation/risk assumptions, and calculation-run records are relational. The initial RAG design uses pgvector in the same PostgreSQL deployment.

PK/FK notation

PK means primary key. FK means foreign key. A notation such as `child.company_id -> companies.company_id` means the child column references the parent key. Multi-column keys are shown in full.

Important implementation boundary

The core schema is a target-state design. The optional trading DDL must not be applied in Release 1. The SQL has been statically inspected for table and FK target consistency; a successful live PostgreSQL migration has not been claimed.

2. ER models and domain relationships

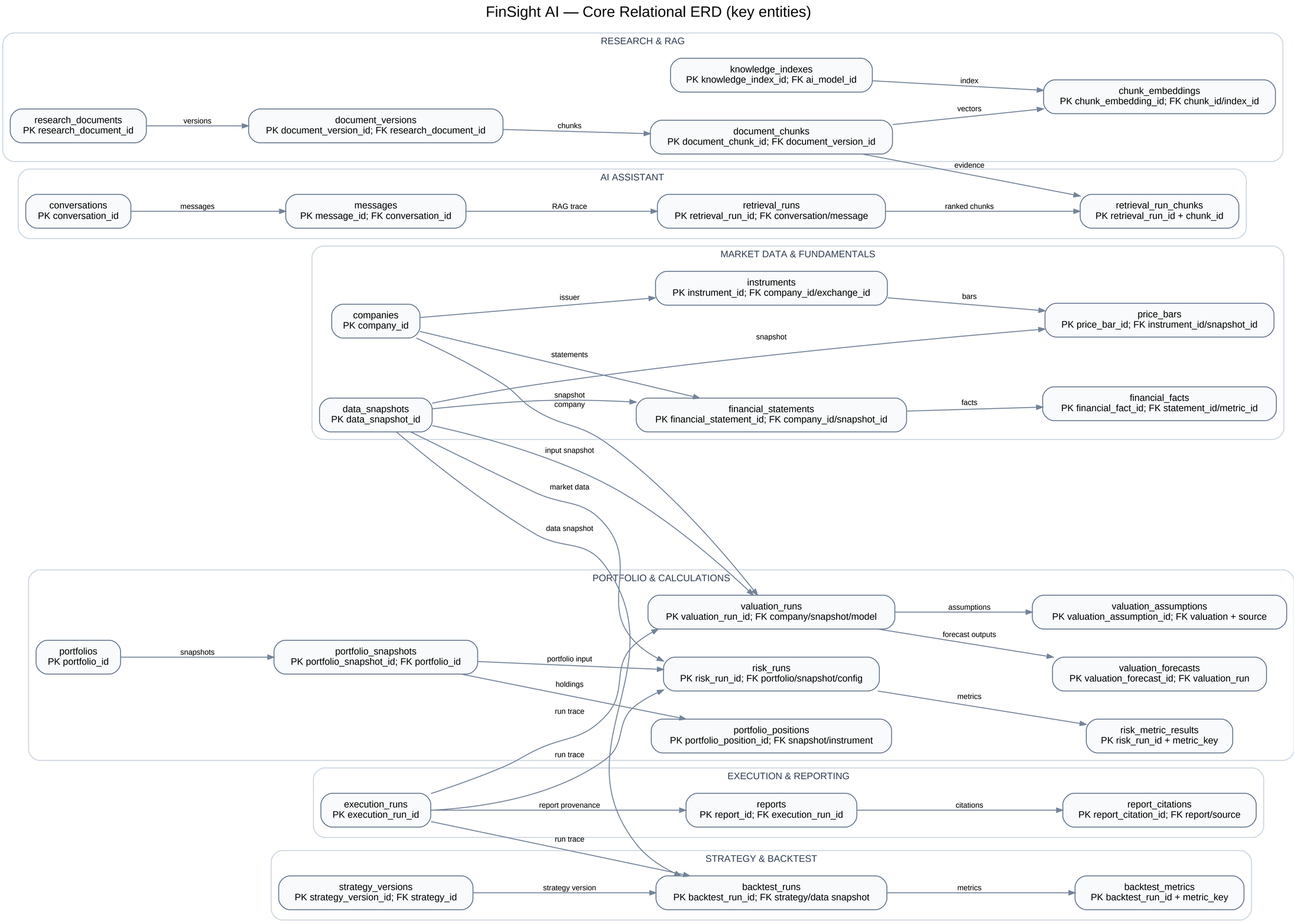
The diagrams below are the models supplied with the design package. Entity-level diagrams show primary and foreign key fields; the table catalogue remains the authoritative compact inventory for the complete declared table/column set.

Diagram index

Model	What it covers	File
System-wide ERD overview	A high-level view of the major connected entities across FinSight AI.	`ERD_overview.svg`
Domain dependency map	A cross-domain workflow/dependency map. It is not a row-level ER diagram.	`ERD_domain_map.svg`
Identity, access and workspace	Users, RBAC, workspace membership, security, settings, ownership and sharing.	`ERD_identity_workspace.svg`
Market data and fundamentals	Companies, instruments, providers, snapshots, price/fundamental records, corporate actions and quality lineage.	`ERD_market_data.svg`
Data explorer and lineage	User datasets, sources, saved queries, transformations, import lineage and exports.	`ERD_data_explorer.svg`
Research library, RAG and AI	Documents, versions, chunks, embeddings, knowledge indexes, conversations, citations and retrieval traces.	`ERD_rag_ai.svg`
Portfolio and dashboard analytics	Watchlists, portfolios, holdings/snapshots/returns, dashboards, widgets and alerts.	`ERD_portfolio_dashboard.svg`
DCF valuation and quantitative risk	Versioned valuation runs and assumptions; risk runs, covariance, drawdowns, stress and simulation outputs.	`ERD_valuation_risk.svg`
Screening, strategies, workflows and backtesting	Saved screens, strategy versions, variants, signals, workflow definitions, approvals, backtests and trade-level outputs.	`ERD_strategy_workflow_backtest.svg`
Optional paper trading and broker integration	Future extension: broker and paper-trading accounts, orders, fills, approvals, risk checks, reconciliation and trade journal.	`ERD_optional_trading.svg`

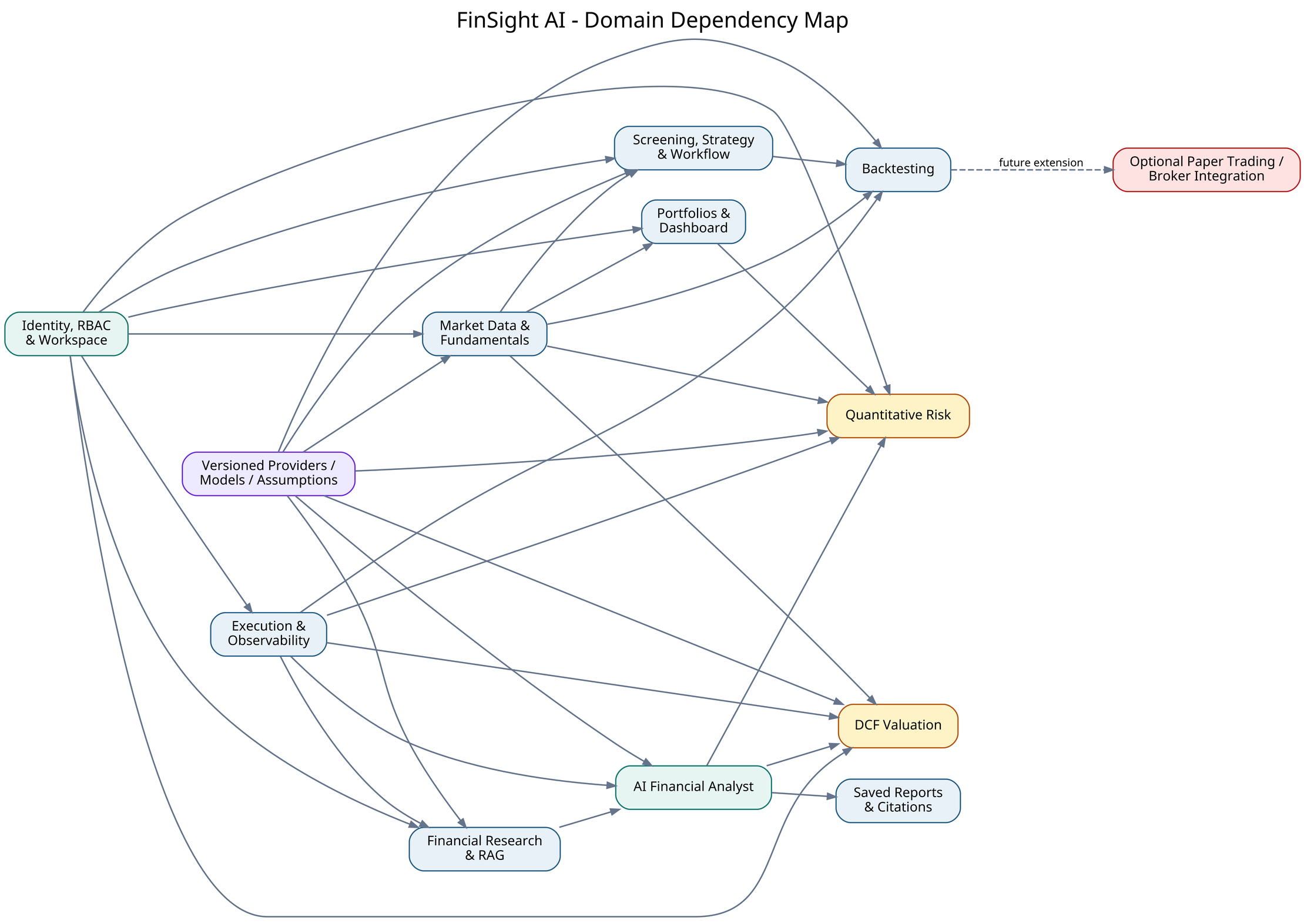
System-wide ERD overview

A high-level view of the major connected entities across FinSight AI.



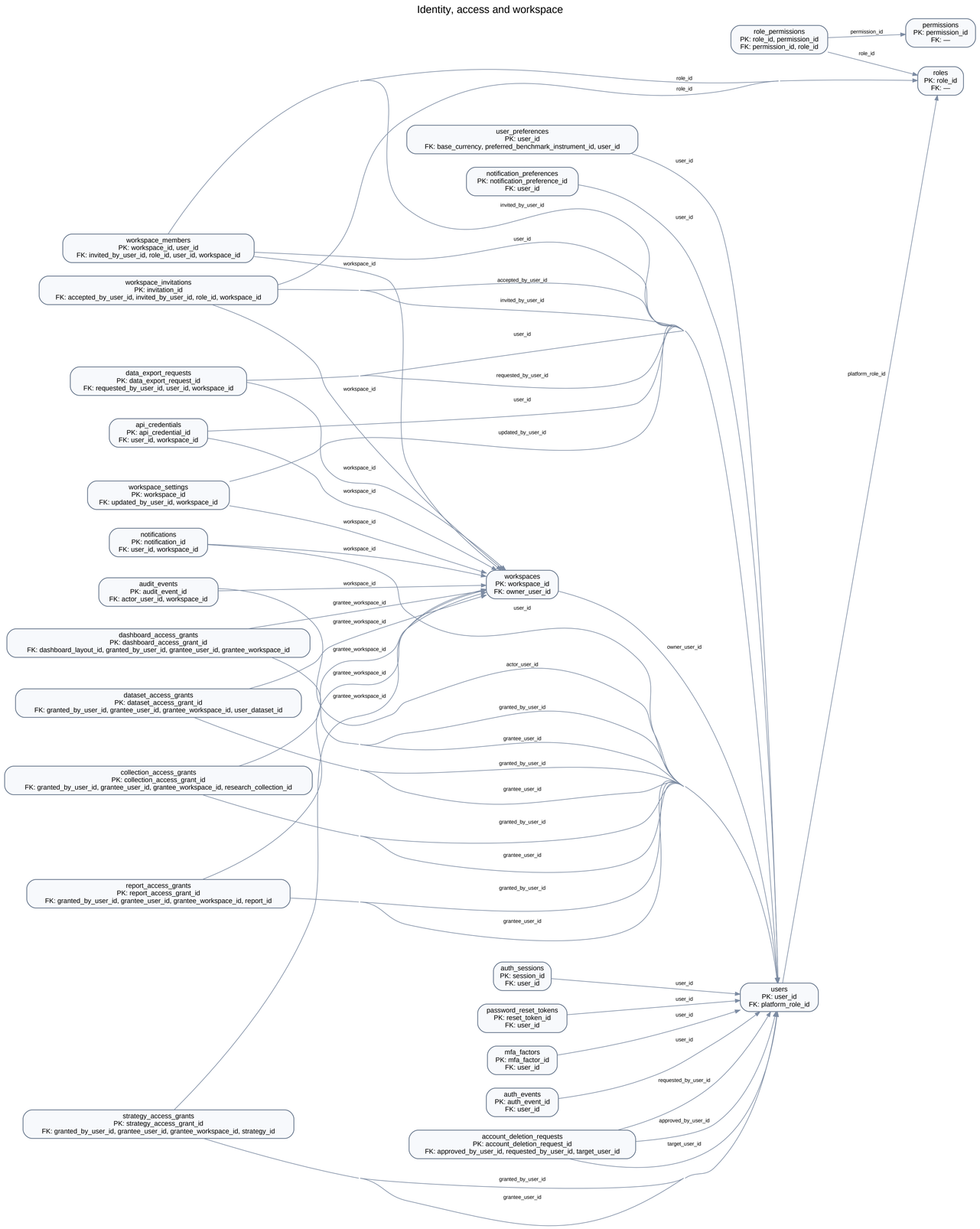
Domain dependency map

A cross-domain workflow/dependency map. It is not a row-level ER diagram.



Identity, access and workspace

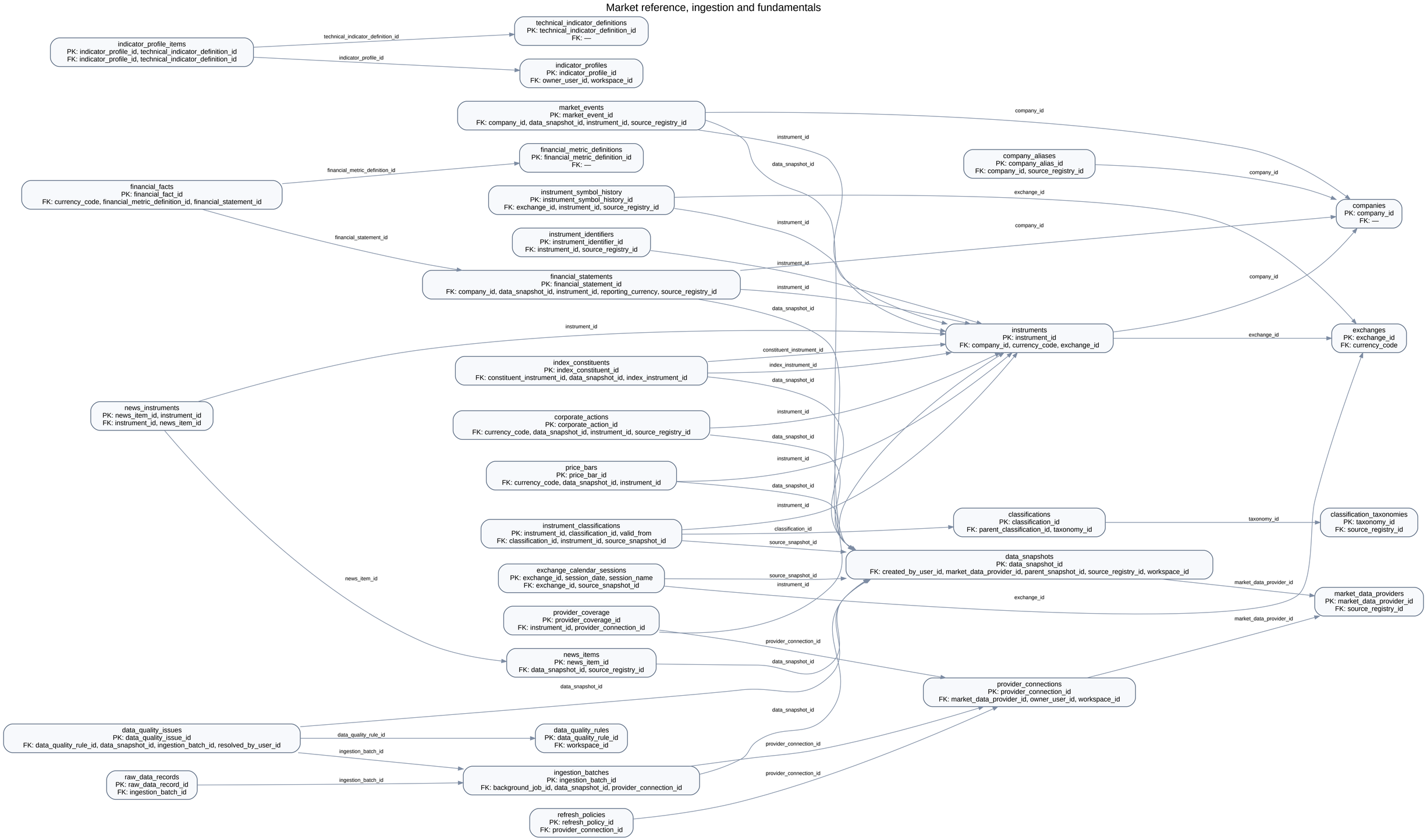
Users, RBAC, workspace membership, security, settings, ownership and sharing.



Market data and fundamentals

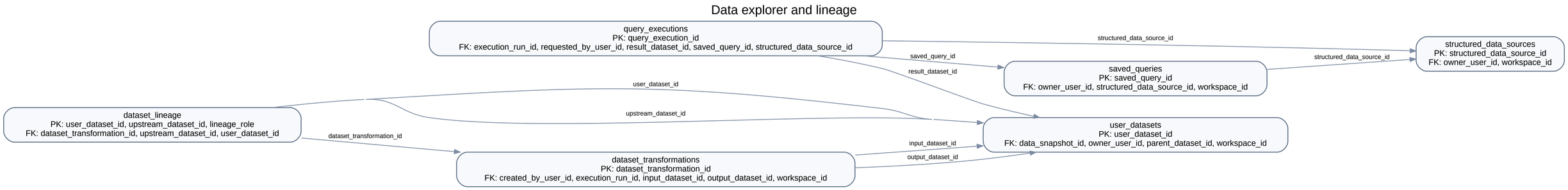
Companies, instruments, providers, snapshots, price/fundamental records, corporate actions and quality lineage.

Market reference, ingestion and fundamentals



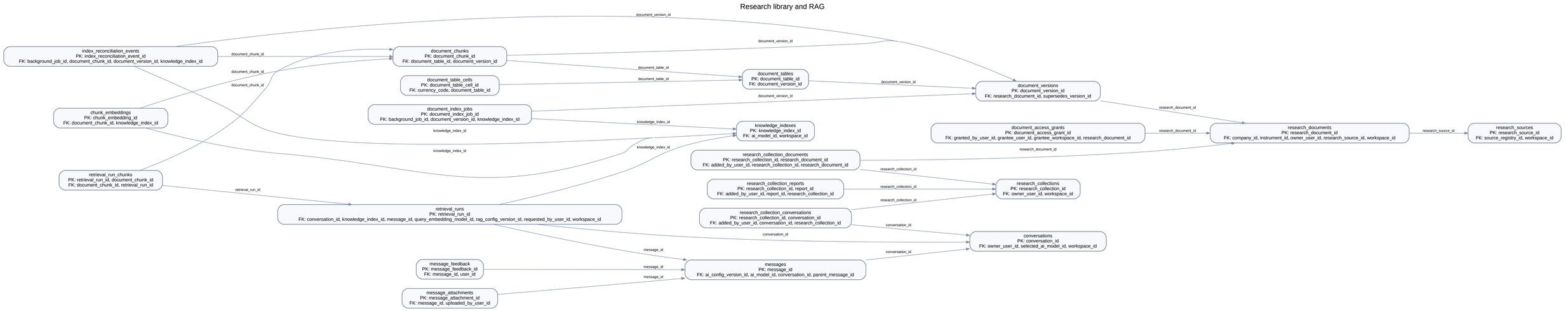
Data explorer and lineage

User datasets, sources, saved queries, transformations, import lineage and exports.



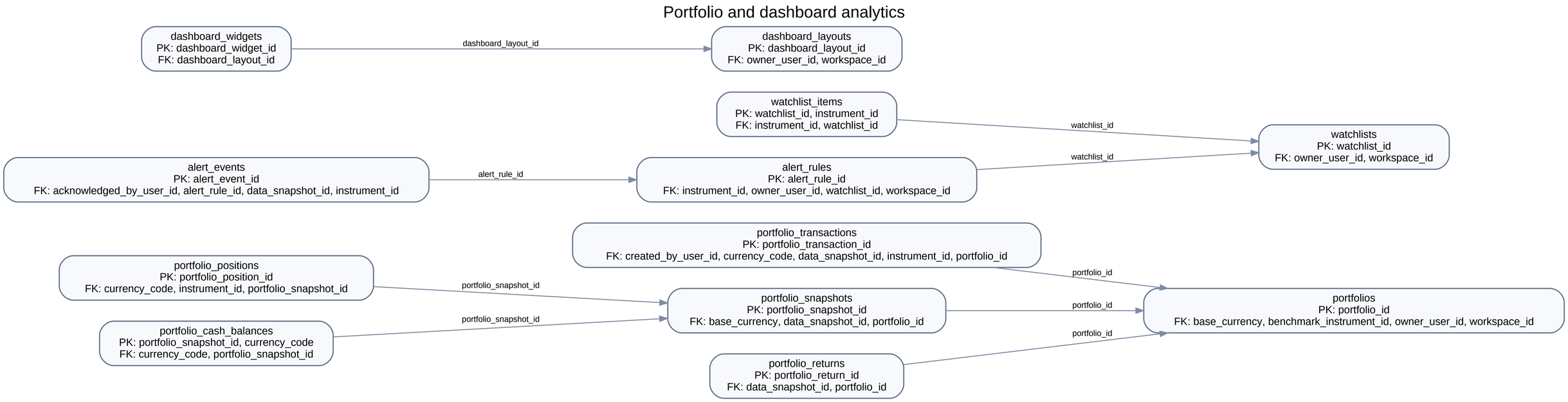
Research library, RAG and AI

Documents, versions, chunks, embeddings, knowledge indexes, conversations, citations and retrieval traces.



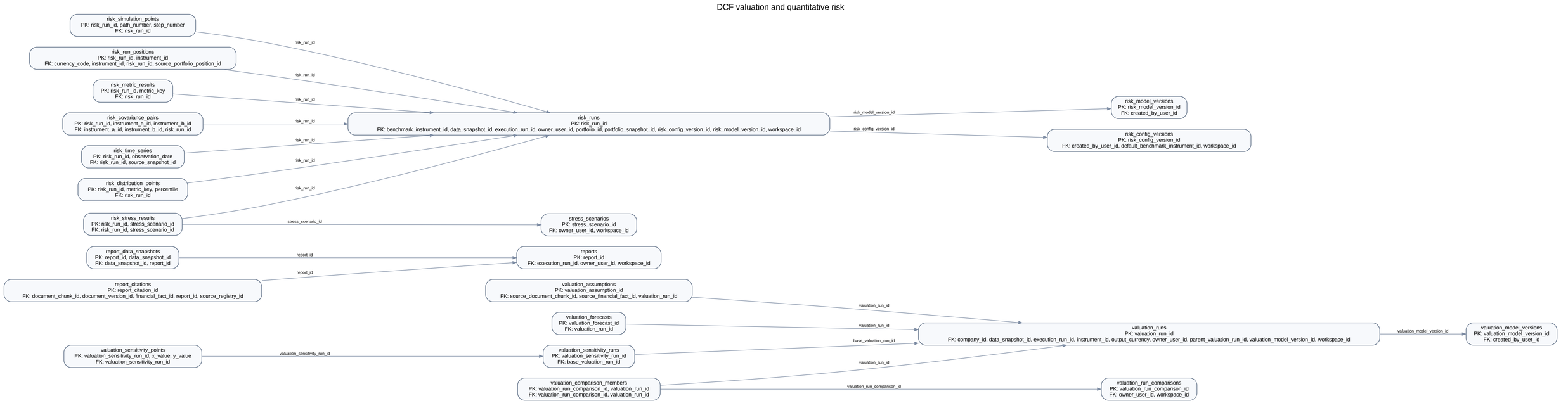
Portfolio and dashboard analytics

Watchlists, portfolios, holdings/snapshots/returns, dashboards, widgets and alerts.



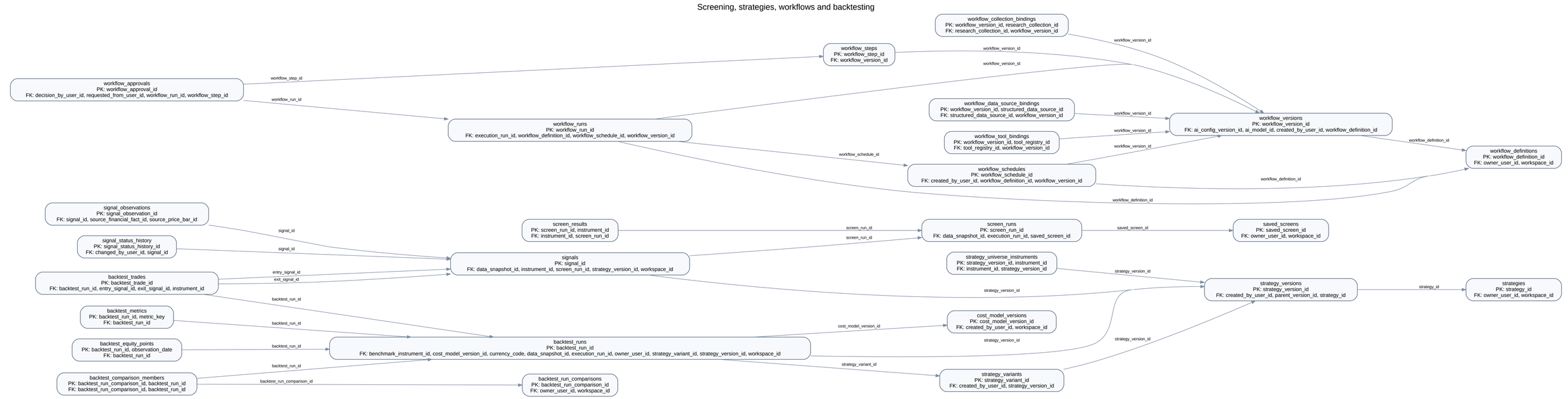
DCF valuation and quantitative risk

Versioned valuation runs and assumptions; risk runs, covariance, drawdowns, stress and simulation outputs.



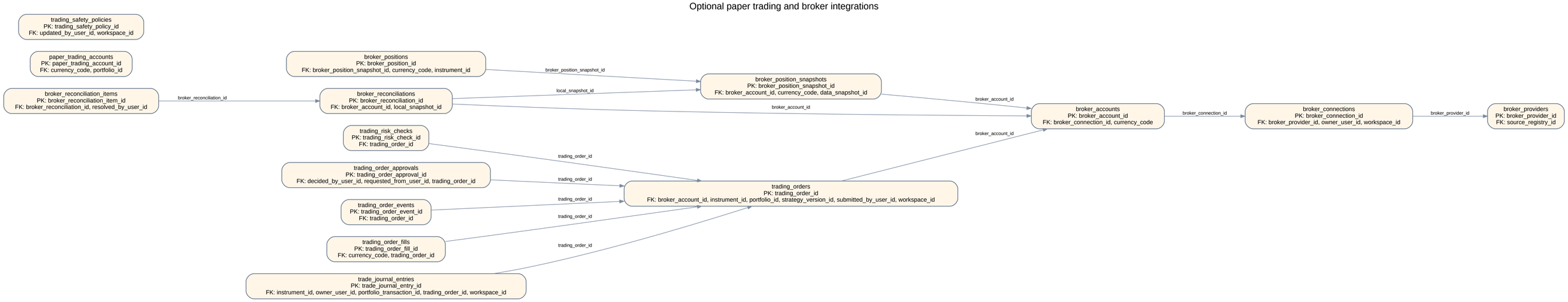
Screening, strategies, workflows and backtesting

Saved screens, strategy versions, variants, signals, workflow definitions, approvals, backtests and trade-level outputs.



Optional paper trading and broker integration

Future extension: broker and paper-trading accounts, orders, fills, approvals, risk checks, reconciliation and trade journal.



3. Complete table schema catalogue

The following sections enumerate all 173 CREATE TABLE definitions. For each table, the reference includes its domain, purpose, declared columns and types, PK/FK markers, nullability/default/unique/check flags, and the FK relationships. Exact SQL defaults, CHECK expressions, indexes, extension declarations and other DDL remain in the accompanying SQL source files.

Domain counts

Domain	Tables	Implementation note
Identity, access and workspace (24 tables)	24	Accounts, workspace boundaries, roles, permissions, preferences and saved-item access.
AI configuration and operations (16 tables)	16	AI providers/models, versioned config, tools, execution runs, jobs and monitoring.
Market reference, ingestion and fundamentals (31 tables)	31	Canonical market entities, snapshots, prices, actions, statements/facts and quality.
Data explorer and lineage (6 tables)	6	Imported datasets, schema/queries, transformations, lineage and exports.
Research library and RAG (21 tables)	21	Source-backed documents, versions, chunking, vector embeddings and retrieval evidence.
Portfolio and dashboard analytics (12 tables)	12	Portfolios, positions, return series, dashboard layout, widgets and watchlists.
DCF valuation and quantitative risk (22 tables)	22	Versioned assumptions, valuation forecasts/sensitivity and reproducible risk outputs.
Screening, strategies, workflows and backtesting (26 tables)	26	Saved screens, strategies, signals, approved workflows and backtest reports.
Optional paper trading and broker integrations (15 tables)	15	Future extension only; do not apply with Release 1 core schema.

Column legend

PK = primary-key column. FK -> parent.column = foreign-key target. NOT NULL = declared required column. DEFAULT = default expression is declared in SQL. UNIQUE = column-level uniqueness. CHECK = a column-level check expression is declared. Composite keys and table-level constraints are listed below the column inventory where applicable.

3.1 Identity, access and workspace (24 tables)

Implementation scope
CORE TARGET SCHEMA - implement in staged releases according to dependencies and priorities.

roles [CORE]

Platform/workspace roles used by RBAC.

PK: `role_id`
FKs: none declared

Column	Type	Declared flags / references
role_id	UUID	PK; DEFAULT
role_key	VARCHAR(50)	NOT NULL; UNIQUE
display_name	VARCHAR(100)	NOT NULL
scope	VARCHAR(20)	NOT NULL; CHECK
description	TEXT	-
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

permissions [CORE]

Named backend permissions.

PK: `permission_id`
FKs: none declared

Column	Type	Declared flags / references
permission_id	UUID	PK; DEFAULT
permission_key	VARCHAR(120)	NOT NULL; UNIQUE
description	TEXT	NOT NULL

role_permissions [CORE]

Many-to-many role-to-permission mapping.

PK: `role_id, permission_id`
FKs: `role_id -> roles.role_id; permission_id -> permissions.permission_id`

Column	Type	Declared flags / references
role_id	UUID	PK; FK -> roles.role_id
permission_id	UUID	PK; FK -> permissions.permission_id

users [CORE]

User identity and account status; passwords are hashes only.

PK: `user_id`
FKs: `platform_role_id -> roles.role_id`

Column	Type	Declared flags / references
user_id	UUID	PK; DEFAULT
email	VARCHAR(320)	NOT NULL
password_hash	TEXT	-
display_name	VARCHAR(160)	NOT NULL
account_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
platform_role_id	UUID	FK -> roles.role_id
email_verified_at	TIMESTAMPTZ	-
mfa_required	BOOLEAN	NOT NULL; DEFAULT
last_login_at	TIMESTAMPTZ	-
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT
updated_at	TIMESTAMPTZ	NOT NULL; DEFAULT
deleted_at	TIMESTAMPTZ	-

workspaces [CORE]

Tenant/security boundary for user-created resources.

PK: workspace_id

FKs: owner_user_id -> users.user_id

Column	Type	Declared flags / references
workspace_id	UUID	PK; DEFAULT
name	VARCHAR(160)	NOT NULL
slug	VARCHAR(180)	NOT NULL; UNIQUE
owner_user_id	UUID	FK -> users.user_id; NOT NULL
workspace_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT
updated_at	TIMESTAMPTZ	NOT NULL; DEFAULT

workspace_members [CORE]

User membership and workspace role.

PK: workspace_id, user_id

FKs: workspace_id -> workspaces.workspace_id; user_id -> users.user_id; role_id -> roles.role_id; invited_by_user_id -> users.user_id

Column	Type	Declared flags / references
workspace_id	UUID	PK; FK -> workspaces.workspace_id
user_id	UUID	PK; FK -> users.user_id
role_id	UUID	FK -> roles.role_id; NOT NULL
member_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
invited_by_user_id	UUID	FK -> users.user_id
joined_at	TIMESTAMPTZ	-
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

workspace_invitations [CORE]

Admin invitations and expiry/revocation.

PK: invitation_id

FKs: workspace_id -> workspaces.workspace_id; role_id -> roles.role_id; invited_by_user_id -> users.user_id; accepted_by_user_id -> users.user_id

Column	Type	Declared flags / references
invitation_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id; NOT NULL
email	VARCHAR(320)	NOT NULL
role_id	UUID	FK -> roles.role_id; NOT NULL
invited_by_user_id	UUID	FK -> users.user_id; NOT NULL
invitation_token_hash	TEXT	NOT NULL; UNIQUE
invitation_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
expires_at	TIMESTAMPTZ	NOT NULL
accepted_by_user_id	UUID	FK -> users.user_id
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

auth_sessions [CORE]

Session and refresh-token lifecycle metadata.

PK: session_id

FKs: user_id -> users.user_id

Column	Type	Declared flags / references
session_id	UUID	PK; DEFAULT
user_id	UUID	FK -> users.user_id; NOT NULL
refresh_token_hash	TEXT	UNIQUE
ip_address	INET	-
user_agent	TEXT	-
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT
last_seen_at	TIMESTAMPTZ	NOT NULL; DEFAULT
expires_at	TIMESTAMPTZ	NOT NULL
revoked_at	TIMESTAMPTZ	-
revoke_reason	VARCHAR(80)	-

password_reset_tokens [CORE]

Hashed single-use password reset tokens.

PK: reset_token_id

FKs: user_id -> users.user_id

Column	Type	Declared flags / references
reset_token_id	UUID	PK; DEFAULT
user_id	UUID	FK -> users.user_id; NOT NULL
token_hash	TEXT	NOT NULL; UNIQUE
expires_at	TIMESTAMPTZ	NOT NULL
consumed_at	TIMESTAMPTZ	-
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

mfa_factors [CORE]

MFA enrollment metadata; secrets stored via secret references.

PK: mfa_factor_id

FKs: user_id -> users.user_id

Column	Type	Declared flags / references
mfa_factor_id	UUID	PK; DEFAULT
user_id	UUID	FK -> users.user_id; NOT NULL
factor_type	VARCHAR(30)	NOT NULL; CHECK
label	VARCHAR(120)	-
secret_ref	TEXT	-
public_key_data	JSONB	-
factor_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
enrolled_at	TIMESTAMPTZ	-
last_used_at	TIMESTAMPTZ	-
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

auth_events [CORE]

Sign-in/security event history.

PK: auth_event_id

FKs: user_id -> users.user_id

Column	Type	Declared flags / references
auth_event_id	UUID	PK; DEFAULT
user_id	UUID	FK -> users.user_id
email_attempted	VARCHAR(320)	-
event_type	VARCHAR(50)	NOT NULL
outcome	VARCHAR(20)	NOT NULL; CHECK
ip_address	INET	-
user_agent	TEXT	-
detail	JSONB	NOT NULL; DEFAULT
occurred_at	TIMESTAMPTZ	NOT NULL; DEFAULT

data_export_requests [CORE]

Tracked user/workspace data export request and artifact lifecycle.

PK: data_export_request_id

FKs: user_id -> users.user_id; workspace_id -> workspaces.workspace_id; requested_by_user_id -> users.user_id

Column	Type	Declared flags / references
data_export_request_id	UUID	PK; DEFAULT
user_id	UUID	FK -> users.user_id; NOT NULL
workspace_id	UUID	FK -> workspaces.workspace_id
requested_by_user_id	UUID	FK -> users.user_id; NOT NULL
export_scope	JSONB	NOT NULL
request_status	VARCHAR(24)	NOT NULL; DEFAULT; CHECK
artifact_uri	TEXT	-
artifact_expires_at	TIMESTAMPTZ	-
requested_at	TIMESTAMPTZ	NOT NULL; DEFAULT
completed_at	TIMESTAMPTZ	-
error_redacted	TEXT	-

account_deletion_requests [CORE]

Deletion request, approval and retention/ownership plan for account data.

PK: account_deletion_request_id

FKs: target_user_id -> users.user_id; requested_by_user_id -> users.user_id; approved_by_user_id -> users.user_id

Column	Type	Declared flags / references
account_deletion_request_id	UUID	PK; DEFAULT
target_user_id	UUID	FK -> users.user_id; NOT NULL
requested_by_user_id	UUID	FK -> users.user_id; NOT NULL
approved_by_user_id	UUID	FK -> users.user_id
request_status	VARCHAR(24)	NOT NULL; DEFAULT; CHECK
retention_and_ownership_plan	JSONB	NOT NULL; DEFAULT
requested_at	TIMESTAMPTZ	NOT NULL; DEFAULT
approved_at	TIMESTAMPTZ	-
completed_at	TIMESTAMPTZ	-
decision_note	TEXT	-

api_credentials [CORE]

Hashed, scoped application API credentials.

PK: `api_credential_id`

FKs: `user_id -> users.user_id; workspace_id -> workspaces.workspace_id`

Column	Type	Declared flags / references
api_credential_id	UUID	PK; DEFAULT
user_id	UUID	FK -> users.user_id; NOT NULL
workspace_id	UUID	FK -> workspaces.workspace_id
label	VARCHAR(120)	NOT NULL
token_prefix	VARCHAR(24)	NOT NULL
token_hash	TEXT	NOT NULL; UNIQUE
scopes	TEXT[]	NOT NULL; DEFAULT
last_used_at	TIMESTAMPTZ	-
expires_at	TIMESTAMPTZ	-
revoked_at	TIMESTAMPTZ	-
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

user_preferences [CORE]

Personal UI, market and display preferences.

PK: `user_id`

FKs: `user_id -> users.user_id`

Column	Type	Declared flags / references
user_id	UUID	PK; FK -> users.user_id
theme	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
language_code	VARCHAR(12)	NOT NULL; DEFAULT
timezone_name	VARCHAR(80)	NOT NULL; DEFAULT
default_market	VARCHAR(20)	NOT NULL; DEFAULT
base_currency	CHAR(3)	NOT NULL; DEFAULT
preferred_benchmark_instrument_id	UUID	-
preferences	JSONB	NOT NULL; DEFAULT
updated_at	TIMESTAMPTZ	NOT NULL; DEFAULT

workspace_settings [CORE]

Workspace policy and retention settings.

PK: `workspace_id`

FKs: `workspace_id -> workspaces.workspace_id; updated_by_user_id -> users.user_id`

Column	Type	Declared flags / references
workspace_id	UUID	PK; FK -> workspaces.workspace_id
settings	JSONB	NOT NULL; DEFAULT
retention_policy	JSONB	NOT NULL; DEFAULT
allow_sharing	BOOLEAN	NOT NULL; DEFAULT
allow_public_registration	BOOLEAN	NOT NULL; DEFAULT
updated_by_user_id	UUID	FK -> users.user_id
updated_at	TIMESTAMPTZ	NOT NULL; DEFAULT

notification_preferences [CORE]

Per-user notification channel controls.

PK: notification_preference_id

FKs: user_id -> users.user_id

Column	Type	Declared flags / references
notification_preference_id	UUID	PK; DEFAULT
user_id	UUID	FK -> users.user_id; NOT NULL
notification_type	VARCHAR(80)	NOT NULL
channel	VARCHAR(30)	NOT NULL; CHECK
enabled	BOOLEAN	NOT NULL; DEFAULT
config	JSONB	NOT NULL; DEFAULT

Table-level constraints:

UNIQUE (user_id, notification_type, channel)

notifications [CORE]

In-app/outbound notification state.

PK: notification_id

FKs: user_id -> users.user_id; workspace_id -> workspaces.workspace_id

Column	Type	Declared flags / references
notification_id	UUID	PK; DEFAULT
user_id	UUID	FK -> users.user_id; NOT NULL
workspace_id	UUID	FK -> workspaces.workspace_id
notification_type	VARCHAR(80)	NOT NULL
title	VARCHAR(240)	NOT NULL
body	TEXT	-
payload	JSONB	NOT NULL; DEFAULT
read_at	TIMESTAMPTZ	-
delivery_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

audit_events [CORE]

Append-only security/admin audit trail.

PK: audit_event_id

FKs: workspace_id -> workspaces.workspace_id; actor_user_id -> users.user_id

Column	Type	Declared flags / references
audit_event_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id
actor_user_id	UUID	FK -> users.user_id
action_key	VARCHAR(120)	NOT NULL
resource_type	VARCHAR(80)	NOT NULL
resource_id	UUID	-
outcome	VARCHAR(20)	NOT NULL; CHECK
correlation_id	UUID	-
details_redacted	JSONB	NOT NULL; DEFAULT
occurred_at	TIMESTAMPTZ	NOT NULL; DEFAULT

dashboard_access_grants [CORE]

Explicit dashboard sharing rights.

PK: dashboard_access_grant_id

FKs: dashboard_layout_id -> dashboard_layouts.dashboard_layout_id; grantee_user_id -> users.user_id; grantee_workspace_id -> workspaces.workspace_id; granted_by_user_id -> users.user_id

Column	Type	Declared flags / references
dashboard_access_grant_id	UUID	PK; DEFAULT
dashboard_layout_id	UUID	FK -> dashboard_layouts.dashboard_layout_id; NOT NULL
grantee_user_id	UUID	FK -> users.user_id
grantee_workspace_id	UUID	FK -> workspaces.workspace_id
permission_level	VARCHAR(20)	NOT NULL; CHECK
granted_by_user_id	UUID	FK -> users.user_id; NOT NULL
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
CHECK (num_nonnulls(grantee_user_id, grantee_workspace_id) = 1)
```

dataset_access_grants [CORE]

Explicit dataset sharing rights.

PK: dataset_access_grant_id

FKs: user_dataset_id -> user_datasets.user_dataset_id; grantee_user_id -> users.user_id; grantee_workspace_id -> workspaces.workspace_id; granted_by_user_id -> users.user_id

Column	Type	Declared flags / references
dataset_access_grant_id	UUID	PK; DEFAULT
user_dataset_id	UUID	FK -> user_datasets.user_dataset_id; NOT NULL
grantee_user_id	UUID	FK -> users.user_id
grantee_workspace_id	UUID	FK -> workspaces.workspace_id
permission_level	VARCHAR(20)	NOT NULL; CHECK
granted_by_user_id	UUID	FK -> users.user_id; NOT NULL
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
CHECK (num_nonnulls(grantee_user_id, grantee_workspace_id) = 1)
```

collection_access_grants [CORE]

Explicit research collection sharing rights.

PK: collection_access_grant_id

FKs: research_collection_id -> research_collections.research_collection_id; grantee_user_id -> users.user_id; grantee_workspace_id -> workspaces.workspace_id; granted_by_user_id -> users.user_id

Column	Type	Declared flags / references
collection_access_grant_id	UUID	PK; DEFAULT
research_collection_id	UUID	FK -> research_collections.research_collection_id; NOT NULL
grantee_user_id	UUID	FK -> users.user_id
grantee_workspace_id	UUID	FK -> workspaces.workspace_id
permission_level	VARCHAR(20)	NOT NULL; CHECK
granted_by_user_id	UUID	FK -> users.user_id; NOT NULL
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
CHECK (num_nonnulls(grantee_user_id, grantee_workspace_id) = 1)
```

report_access_grants [CORE]

Explicit saved report sharing rights.

PK: report_access_grant_id

FKs: report_id -> reports.report_id; grantee_user_id -> users.user_id; grantee_workspace_id -> workspaces.workspace_id; granted_by_user_id -> users.user_id

Column	Type	Declared flags / references
report_access_grant_id	UUID	PK; DEFAULT
report_id	UUID	FK -> reports.report_id; NOT NULL
grantee_user_id	UUID	FK -> users.user_id
grantee_workspace_id	UUID	FK -> workspaces.workspace_id
permission_level	VARCHAR(20)	NOT NULL; CHECK
granted_by_user_id	UUID	FK -> users.user_id; NOT NULL
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
CHECK (num_nonnulls(grantee_user_id, grantee_workspace_id) = 1)
```

strategy_access_grants [CORE]

Explicit strategy sharing rights.

PK: strategy_access_grant_id

FKs: strategy_id -> strategies.strategy_id; grantee_user_id -> users.user_id; grantee_workspace_id -> workspaces.workspace_id; granted_by_user_id -> users.user_id

Column	Type	Declared flags / references
strategy_access_grant_id	UUID	PK; DEFAULT
strategy_id	UUID	FK -> strategies.strategy_id; NOT NULL
grantee_user_id	UUID	FK -> users.user_id
grantee_workspace_id	UUID	FK -> workspaces.workspace_id
permission_level	VARCHAR(20)	NOT NULL; CHECK
granted_by_user_id	UUID	FK -> users.user_id; NOT NULL
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
CHECK (num_nonnulls(grantee_user_id, grantee_workspace_id) = 1)
```

3.2 AI configuration and operations (16 tables)

Implementation scope
CORE TARGET SCHEMA - implement in staged releases according to dependencies and priorities.

currencies [CORE]

Supported ISO currency reference codes and minor units.

PK: currency_code
FKs: none declared

Column	Type	Declared flags / references
currency_code	CHAR(3)	PK
currency_name	VARCHAR(80)	NOT NULL
minor_unit	SMALLINT	NOT NULL; DEFAULT; CHECK

source_registry [CORE]

Canonical data/research source, licensing, attribution and permitted use.

PK: source_registry_id
FKs: created_by_user_id -> users.user_id

Column	Type	Declared flags / references
source_registry_id	UUID	PK; DEFAULT
source_key	VARCHAR(160)	NOT NULL; UNIQUE
source_type	VARCHAR(30)	NOT NULL; CHECK
display_name	VARCHAR(200)	NOT NULL
publisher	VARCHAR(200)	-
base_url	TEXT	-
licence_status	VARCHAR(30)	NOT NULL; DEFAULT; CHECK
licence_url	TEXT	-
permitted_usage	JSONB	NOT NULL; DEFAULT
attribution_text	TEXT	-
terms_reviewed_at	TIMESTAMPTZ	-
created_by_user_id	UUID	FK -> users.user_id
enabled	BOOLEAN	NOT NULL; DEFAULT
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

ai_providers [CORE]

Provider interface configuration and health.

PK: ai_provider_id
FKs: none declared

Column	Type	Declared flags / references
ai_provider_id	UUID	PK; DEFAULT
provider_key	VARCHAR(100)	NOT NULL; UNIQUE
display_name	VARCHAR(160)	NOT NULL
provider_kind	VARCHAR(30)	NOT NULL; CHECK
base_url	TEXT	-
secret_ref	TEXT	-
provider_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
last_health_check_at	TIMESTAMPTZ	-
non_secret_config	JSONB	NOT NULL; DEFAULT
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

ai_models [CORE]

Discoverable chat, embedding and reranking models.

PK: ai_model_id

FKs: ai_provider_id -> ai_providers.ai_provider_id

Column	Type	Declared flags / references
ai_model_id	UUID	PK; DEFAULT
ai_provider_id	UUID	FK -> ai_providers.ai_provider_id; NOT NULL
model_key	VARCHAR(160)	NOT NULL
model_name	VARCHAR(240)	NOT NULL
model_type	VARCHAR(24)	NOT NULL; CHECK
embedding_dimension	INTEGER	CHECK
context_window_tokens	INTEGER	CHECK
is_available	BOOLEAN	NOT NULL; DEFAULT
model_metadata	JSONB	NOT NULL; DEFAULT
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (ai_provider_id, model_key)
CHECK ((model_type = 'embedding' AND embedding_dimension IS NOT NULL) OR model_type <> 'embedding')
```

ai_config_versions [CORE]

Versioned inference parameters; invocations point to the version used.

PK: ai_config_version_id

FKs: ai_model_id -> ai_models.ai_model_id; created_by_user_id -> users.user_id

Column	Type	Declared flags / references
ai_config_version_id	UUID	PK; DEFAULT
ai_model_id	UUID	FK -> ai_models.ai_model_id; NOT NULL
version_number	INTEGER	NOT NULL; CHECK
config_name	VARCHAR(160)	NOT NULL
generation_parameters	JSONB	NOT NULL; DEFAULT
is_active	BOOLEAN	NOT NULL; DEFAULT
created_by_user_id	UUID	FK -> users.user_id
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (ai_model_id, version_number)
```

rag_config_versions [CORE]

Versioned chunking/retrieval/security configuration.

PK: rag_config_version_id

FKs: workspace_id -> workspaces.workspace_id; created_by_user_id -> users.user_id

Column	Type	Declared flags / references
rag_config_version_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id
version_number	INTEGER	NOT NULL; CHECK
config_name	VARCHAR(160)	NOT NULL
chunking_config	JSONB	NOT NULL; DEFAULT
retrieval_config	JSONB	NOT NULL; DEFAULT
permission_filter_policy	JSONB	NOT NULL; DEFAULT
is_active	BOOLEAN	NOT NULL; DEFAULT
created_by_user_id	UUID	FK -> users.user_id
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (workspace_id, version_number)
```

tool_registry [CORE]

Allowlisted backend tools and permission scopes.

PK: `tool_registry_id`

FKs: none declared

Column	Type	Declared flags / references
tool_registry_id	UUID	PK; DEFAULT
tool_key	VARCHAR(120)	NOT NULL; UNIQUE
display_name	VARCHAR(160)	NOT NULL
description	TEXT	-
input_schema	JSONB	NOT NULL; DEFAULT
output_schema	JSONB	NOT NULL; DEFAULT
permission_scope	VARCHAR(100)	NOT NULL
enabled	BOOLEAN	NOT NULL; DEFAULT
high_impact	BOOLEAN	NOT NULL; DEFAULT
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

workspace_tool_policies [CORE]

Workspace-level allow/deny and operation mode for AI tools.

PK: `workspace_id, tool_registry_id`

FKs: `workspace_id -> workspaces.workspace_id; tool_registry_id -> tool_registry.tool_registry_id; updated_by_user_id -> users.user_id`

Column	Type	Declared flags / references
workspace_id	UUID	PK; FK -> workspaces.workspace_id
tool_registry_id	UUID	PK; FK -> tool_registry.tool_registry_id
enabled	BOOLEAN	NOT NULL; DEFAULT
permission_mode	VARCHAR(24)	NOT NULL; DEFAULT; CHECK
policy_config	JSONB	NOT NULL; DEFAULT
updated_by_user_id	UUID	FK -> users.user_id
updated_at	TIMESTAMPTZ	NOT NULL; DEFAULT

execution_runs [CORE]

Stable parent record for long-running work/calculations.

PK: `execution_run_id`

FKs: `workspace_id -> workspaces.workspace_id; initiated_by_user_id -> users.user_id`

Column	Type	Declared flags / references
execution_run_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id
initiated_by_user_id	UUID	FK -> users.user_id
execution_type	VARCHAR(40)	NOT NULL
status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
idempotency_key	VARCHAR(200)	-
correlation_id	UUID	-
input_parameters	JSONB	NOT NULL; DEFAULT
output_summary	JSONB	NOT NULL; DEFAULT
output_artifact_uri	TEXT	-
error_code	VARCHAR(100)	-
error_message_redacted	TEXT	-
queued_at	TIMESTAMPTZ	NOT NULL; DEFAULT
started_at	TIMESTAMPTZ	-
finished_at	TIMESTAMPTZ	-
duration_ms	BIGINT	CHECK

Table-level constraints:

```
UNIQUE (workspace_id, idempotency_key)
```

execution_steps [CORE]

Ordered execution trace steps.

PK: execution_step_id

FKs: execution_run_id -> execution_runs.execution_run_id

Column	Type	Declared flags / references
execution_step_id	UUID	PK; DEFAULT
execution_run_id	UUID	FK -> execution_runs.execution_run_id; NOT NULL
step_number	INTEGER	NOT NULL; CHECK
step_key	VARCHAR(120)	NOT NULL
status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
input_summary	JSONB	NOT NULL; DEFAULT
output_summary	JSONB	NOT NULL; DEFAULT
error_code	VARCHAR(100)	-
started_at	TIMESTAMPTZ	-
finished_at	TIMESTAMPTZ	-

Table-level constraints:

```
UNIQUE (execution_run_id, step_number)
```

execution_events [CORE]

Timestamped operational events for a run.

PK: execution_event_id

FKs: execution_run_id -> execution_runs.execution_run_id; execution_step_id -> execution_steps.execution_step_id

Column	Type	Declared flags / references
execution_event_id	UUID	PK; DEFAULT
execution_run_id	UUID	FK -> execution_runs.execution_run_id; NOT NULL
execution_step_id	UUID	FK -> execution_steps.execution_step_id
event_type	VARCHAR(80)	NOT NULL
severity	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
message_redacted	TEXT	-
event_data	JSONB	NOT NULL; DEFAULT
occurred_at	TIMESTAMPTZ	NOT NULL; DEFAULT

service_health_checks [CORE]

Recent status/latency for dependency health dashboard.

PK: service_health_check_id

FKs: none declared

Column	Type	Declared flags / references
service_health_check_id	UUID	PK; DEFAULT
service_key	VARCHAR(120)	NOT NULL
service_instance	VARCHAR(240)	-
status	VARCHAR(24)	NOT NULL; CHECK
latency_ms	BIGINT	CHECK
details_redacted	JSONB	NOT NULL; DEFAULT

Table-level constraints:

```
checked_at TIMESTAMPTZ NOT NULL DEFAULT now()
```

background_jobs [CORE]

Retryable asynchronous work with idempotency key.

PK: background_job_id

FKs: execution_run_id -> execution_runs.execution_run_id; workspace_id -> workspaces.workspace_id; requested_by_user_id -> users.user_id

Column	Type	Declared flags / references
background_job_id	UUID	PK; DEFAULT
execution_run_id	UUID	FK -> execution_runs.execution_run_id
workspace_id	UUID	FK -> workspaces.workspace_id
requested_by_user_id	UUID	FK -> users.user_id
job_type	VARCHAR(100)	NOT NULL
status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
idempotency_key	VARCHAR(200)	UNIQUE
payload	JSONB	NOT NULL; DEFAULT
max_attempts	SMALLINT	NOT NULL; DEFAULT; CHECK
attempt_count	SMALLINT	NOT NULL; DEFAULT; CHECK
available_at	TIMESTAMPTZ	NOT NULL; DEFAULT
locked_at	TIMESTAMPTZ	-
started_at	TIMESTAMPTZ	-
finished_at	TIMESTAMPTZ	-
last_error_redacted	TEXT	-
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

job_attempts [CORE]

Attempt-level outcomes for background jobs.

PK: job_attempt_id

FKs: background_job_id -> background_jobs.background_job_id

Column	Type	Declared flags / references
job_attempt_id	UUID	PK; DEFAULT
background_job_id	UUID	FK -> background_jobs.background_job_id; NOT NULL
attempt_number	SMALLINT	NOT NULL; CHECK
worker_id	VARCHAR(120)	-
status	VARCHAR(20)	NOT NULL; CHECK
started_at	TIMESTAMPTZ	NOT NULL; DEFAULT
finished_at	TIMESTAMPTZ	-
error_code	VARCHAR(100)	-
error_message_redacted	TEXT	-

Table-level constraints:

```
UNIQUE (background_job_id, attempt_number)
```

model_invocations [CORE]

Model/version/token/latency trace per AI operation.

PK: `model_invocation_id`

FKs: `execution_run_id -> execution_runs.execution_run_id; conversation_id -> conversations.conversation_id; message_id -> messages.message_id; ai_model_id -> ai_models.ai_model_id; ai_config_version_id -> ai_config_versions.ai_config_version_id`

Column	Type	Declared flags / references
model_invocation_id	UUID	PK; DEFAULT
execution_run_id	UUID	FK -> execution_runs.execution_run_id
conversation_id	UUID	FK -> conversations.conversation_id
message_id	UUID	FK -> messages.message_id
ai_model_id	UUID	FK -> ai_models.ai_model_id; NOT NULL
ai_config_version_id	UUID	FK -> ai_config_versions.ai_config_version_id
operation_type	VARCHAR(40)	NOT NULL; CHECK
status	VARCHAR(20)	NOT NULL; CHECK
input_tokens	INTEGER	CHECK
output_tokens	INTEGER	CHECK
elapsed_ms	BIGINT	CHECK
reported_cost	NUMERIC(18,8)	CHECK
error_code	VARCHAR(100)	-
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

tool_invocations [CORE]

Authorized tool call inputs, outcomes and approval status.

PK: `tool_invocation_id`

FKs: `tool_registry_id -> tool_registry.tool_registry_id; message_id -> messages.message_id; execution_run_id -> execution_runs.execution_run_id; requested_by_user_id -> users.user_id`

Column	Type	Declared flags / references
tool_invocation_id	UUID	PK; DEFAULT
tool_registry_id	UUID	FK -> tool_registry.tool_registry_id; NOT NULL
message_id	UUID	FK -> messages.message_id
execution_run_id	UUID	FK -> execution_runs.execution_run_id
requested_by_user_id	UUID	FK -> users.user_id
input_payload	JSONB	NOT NULL; DEFAULT
output_summary	JSONB	NOT NULL; DEFAULT
status	VARCHAR(20)	NOT NULL; CHECK
approval_required	BOOLEAN	NOT NULL; DEFAULT
started_at	TIMESTAMPTZ	-
finished_at	TIMESTAMPTZ	-
error_code	VARCHAR(100)	-

3.3 Market reference, ingestion and fundamentals (31 tables)

Implementation scope
CORE TARGET SCHEMA - implement in staged releases according to dependencies and priorities.

exchanges [CORE]

Exchange identity, timezone, currency and status.

PK: exchange_id
FKs: currency_code -> currencies.currency_code

Column	Type	Declared flags / references
exchange_id	UUID	PK; DEFAULT
mic_code	VARCHAR(12)	UNIQUE
exchange_code	VARCHAR(30)	NOT NULL; UNIQUE
exchange_name	VARCHAR(160)	NOT NULL
country_code	CHAR(2)	-
timezone_name	VARCHAR(80)	NOT NULL
currency_code	CHAR(3)	FK -> currencies.currency_code
is_active	BOOLEAN	NOT NULL; DEFAULT
metadata	JSONB	NOT NULL; DEFAULT

exchange_calendar_sessions [CORE]

Per-exchange session hours and holidays.

PK: exchange_id, session_date, session_name
FKs: exchange_id -> exchanges.exchange_id

Column	Type	Declared flags / references
exchange_id	UUID	PK; FK -> exchanges.exchange_id
session_date	DATE	PK
is_trading_day	BOOLEAN	NOT NULL
market_open_at	TIMESTAMPTZ	-
market_close_at	TIMESTAMPTZ	-
pre_market_open_at	TIMESTAMPTZ	-
post_market_close_at	TIMESTAMPTZ	-
session_name	VARCHAR(40)	PK; DEFAULT
source_snapshot_id	UUID	-

Table-level constraints:
CHECK ((is_trading_day = false) OR (market_open_at IS NOT NULL AND market_close_at IS NOT NULL))

companies [CORE]

Canonical issuer/company identity.

PK: company_id
FKs: none declared

Column	Type	Declared flags / references
company_id	UUID	PK; DEFAULT
legal_name	VARCHAR(240)	NOT NULL
display_name	VARCHAR(240)	-
country_code	CHAR(2)	-
cik	VARCHAR(20)	-
lei	VARCHAR(20)	-
website_url	TEXT	-
incorporation_date	DATE	-
company_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT
updated_at	TIMESTAMPTZ	NOT NULL; DEFAULT

company_aliases [CORE]

Alternative/former names with validity and source.

PK: company_alias_id

FKs: company_id -> companies.company_id; source_registry_id -> source_registry.source_registry_id

Column	Type	Declared flags / references
company_alias_id	UUID	PK; DEFAULT
company_id	UUID	FK -> companies.company_id; NOT NULL
alias_text	VARCHAR(240)	NOT NULL
alias_type	VARCHAR(40)	NOT NULL; CHECK
valid_from	DATE	-
valid_to	DATE	-
source_registry_id	UUID	FK -> source_registry.source_registry_id
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
CHECK (valid_to IS NULL OR valid_from IS NULL OR valid_to >= valid_from)
```

instruments [CORE]

Canonical tradable/trackable instruments including equities and indices.

PK: instrument_id

FKs: company_id -> companies.company_id; exchange_id -> exchanges.exchange_id; currency_code -> currencies.currency_code

Column	Type	Declared flags / references
instrument_id	UUID	PK; DEFAULT
company_id	UUID	FK -> companies.company_id
exchange_id	UUID	FK -> exchanges.exchange_id
symbol	VARCHAR(40)	NOT NULL
instrument_name	VARCHAR(240)	NOT NULL
instrument_type	VARCHAR(30)	NOT NULL; CHECK
currency_code	CHAR(3)	FK -> currencies.currency_code; NOT NULL
isin	VARCHAR(20)	-
figi	VARCHAR(24)	-
listing_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
valid_from	DATE	-
valid_to	DATE	-
metadata	JSONB	NOT NULL; DEFAULT
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT
updated_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
CHECK (valid_to IS NULL OR valid_from IS NULL OR valid_to >= valid_from)
```

instrument_identifiers [CORE]

ISIN/FIGI/vendor and historical identifiers.

PK: instrument_identifier_id

FKs: instrument_id -> instruments.instrument_id; source_registry_id -> source_registry.source_registry_id

Column	Type	Declared flags / references
instrument_identifier_id	UUID	PK; DEFAULT
instrument_id	UUID	FK -> instruments.instrument_id; NOT NULL
identifier_scheme	VARCHAR(40)	NOT NULL
identifier_value	VARCHAR(200)	NOT NULL
valid_from	DATE	-
valid_to	DATE	-
source_registry_id	UUID	FK -> source_registry.source_registry_id

Table-level constraints:

```
UNIQUE (identifier_scheme, identifier_value, valid_from)
CHECK (valid_to IS NULL OR valid_from IS NULL OR valid_to >= valid_from)
```

instrument_symbol_history [CORE]

Historical symbols/tickers with effective dates and source provenance.

PK: instrument_symbol_history_id

FKs: instrument_id -> instruments.instrument_id; exchange_id -> exchanges.exchange_id; source_registry_id -> source_registry.source_registry_id

Column	Type	Declared flags / references
instrument_symbol_history_id	UUID	PK; DEFAULT
instrument_id	UUID	FK -> instruments.instrument_id; NOT NULL
exchange_id	UUID	FK -> exchanges.exchange_id
symbol	VARCHAR(40)	NOT NULL
valid_from	DATE	NOT NULL
valid_to	DATE	-
source_registry_id	UUID	FK -> source_registry.source_registry_id
recorded_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
CHECK (valid_to IS NULL OR valid_to >= valid_from)
UNIQUE (instrument_id, symbol, valid_from)
```

classification_taxonomies [CORE]

Sector/industry/classification systems.

PK: taxonomy_id

FKs: source_registry_id -> source_registry.source_registry_id

Column	Type	Declared flags / references
taxonomy_id	UUID	PK; DEFAULT
taxonomy_key	VARCHAR(80)	NOT NULL; UNIQUE
taxonomy_name	VARCHAR(160)	NOT NULL
source_registry_id	UUID	FK -> source_registry.source_registry_id

classifications [CORE]

Hierarchical sector/industry values.

PK: classification_id

FKs: taxonomy_id -> classification_taxonomies.taxonomy_id; parent_classification_id -> classifications.classification_id

Column	Type	Declared flags / references
classification_id	UUID	PK; DEFAULT
taxonomy_id	UUID	FK -> classification_taxonomies.taxonomy_id; NOT NULL
parent_classification_id	UUID	FK -> classifications.classification_id
classification_code	VARCHAR(80)	-
classification_name	VARCHAR(160)	NOT NULL

Table-level constraints:

```
UNIQUE (taxonomy_id, classification_code)
```

instrument_classifications [CORE]

Time-aware instrument-to-sector/industry membership.

PK: instrument_id, classification_id, valid_from

FKs: instrument_id -> instruments.instrument_id; classification_id -> classifications.classification_id

Column	Type	Declared flags / references
instrument_id	UUID	PK; FK -> instruments.instrument_id
classification_id	UUID	PK; FK -> classifications.classification_id
valid_from	DATE	PK
valid_to	DATE	-
source_snapshot_id	UUID	-

Table-level constraints:

```
CHECK (valid_to IS NULL OR valid_from IS NULL OR valid_to >= valid_from)
```

market_data_providers [CORE]

Provider adapter, regional coverage, delay type and rate limits.

PK: `market_data_provider_id`

FKs: `source_registry_id -> source_registry.source_registry_id`

Column	Type	Declared flags / references
market_data_provider_id	UUID	PK; DEFAULT
source_registry_id	UUID	FK -> source_registry.source_registry_id; NOT NULL
provider_key	VARCHAR(100)	NOT NULL; UNIQUE
display_name	VARCHAR(160)	NOT NULL
supported_regions	TEXT[]	NOT NULL; DEFAULT
data_delay_type	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
adapter_name	VARCHAR(120)	NOT NULL
rate_limit_config	JSONB	NOT NULL; DEFAULT
provider_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
last_health_check_at	TIMESTAMPTZ	-
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

provider_connections [CORE]

Workspace/user-owned provider access; credential stored externally.

PK: `provider_connection_id`

FKs: `market_data_provider_id -> market_data_providers.market_data_provider_id; workspace_id -> workspaces.workspace_id; owner_user_id -> users.user_id`

Column	Type	Declared flags / references
provider_connection_id	UUID	PK; DEFAULT
market_data_provider_id	UUID	FK -> market_data_providers.market_data_provider_id; NOT NULL
workspace_id	UUID	FK -> workspaces.workspace_id; NOT NULL
owner_user_id	UUID	FK -> users.user_id; NOT NULL
connection_name	VARCHAR(160)	NOT NULL
secret_ref	TEXT	-
access_mode	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
connection_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
last_tested_at	TIMESTAMPTZ	-
last_successful_sync_at	TIMESTAMPTZ	-
non_secret_config	JSONB	NOT NULL; DEFAULT
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (provider_connection_id, workspace_id)
```

data_snapshots [CORE]

Versioned, hashed market/fundamental/dataset snapshots and freshness mode.

PK: data_snapshot_id

FKs: workspace_id -> workspaces.workspace_id; market_data_provider_id -> market_data_providers.market_data_provider_id; source_registry_id -> source_registry.source_registry_id; parent_snapshot_id -> data_snapshots.data_snapshot_id; created_by_user_id -> users.user_id

Column	Type	Declared flags / references
data_snapshot_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id
market_data_provider_id	UUID	FK -> market_data_providers.market_data_provider_id
source_registry_id	UUID	FK -> source_registry.source_registry_id
parent_snapshot_id	UUID	FK -> data_snapshots.data_snapshot_id
dataset_type	VARCHAR(60)	NOT NULL
snapshot_key	VARCHAR(240)	-
data_mode	VARCHAR(24)	NOT NULL; CHECK
freshness_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
snapshot_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
as_of_at	TIMESTAMPTZ	-
captured_at	TIMESTAMPTZ	NOT NULL; DEFAULT
coverage_start_at	TIMESTAMPTZ	-
coverage_end_at	TIMESTAMPTZ	-
source_version	VARCHAR(160)	-
content_hash	CHAR(64)	-
record_count	BIGINT	CHECK
quality_summary	JSONB	NOT NULL; DEFAULT
created_by_user_id	UUID	FK -> users.user_id

Table-level constraints:

CHECK (coverage_end_at IS NULL OR coverage_start_at IS NULL OR coverage_end_at >= coverage_start_at)

provider_coverage [CORE]

Fields, intervals and historical coverage/entitlements.

PK: provider_coverage_id

FKs: provider_connection_id -> provider_connections.provider_connection_id; instrument_id -> instruments.instrument_id

Column	Type	Declared flags / references
provider_coverage_id	UUID	PK; DEFAULT
provider_connection_id	UUID	FK -> provider_connections.provider_connection_id; NOT NULL
instrument_id	UUID	FK -> instruments.instrument_id
data_category	VARCHAR(60)	NOT NULL
frequency	VARCHAR(30)	-
field_list	TEXT[]	NOT NULL; DEFAULT
coverage_start_at	TIMESTAMPTZ	-
coverage_end_at	TIMESTAMPTZ	-
entitlement_notes	TEXT	-
last_checked_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

UNIQUE (provider_connection_id, instrument_id, data_category, frequency)

ingestion_batches [CORE]

Idempotent source requests, coverage and accepted/rejected row counts.

PK: `ingestion_batch_id`

FKs: `provider_connection_id -> provider_connections.provider_connection_id; data_snapshot_id -> data_snapshots.data_snapshot_id; background_job_id -> background_jobs.background_job_id`

Column	Type	Declared flags / references
ingestion_batch_id	UUID	PK; DEFAULT
provider_connection_id	UUID	FK -> provider_connections.provider_connection_id
data_snapshot_id	UUID	FK -> data_snapshots.data_snapshot_id
background_job_id	UUID	FK -> background_jobs.background_job_id
batch_type	VARCHAR(60)	NOT NULL
idempotency_key	VARCHAR(200)	NOT NULL; UNIQUE
request_metadata	JSONB	NOT NULL; DEFAULT
status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
requested_start_at	TIMESTAMPTZ	-
requested_end_at	TIMESTAMPTZ	-
rows_received	BIGINT	NOT NULL; DEFAULT; CHECK
rows_accepted	BIGINT	NOT NULL; DEFAULT; CHECK
rows_rejected	BIGINT	NOT NULL; DEFAULT; CHECK
started_at	TIMESTAMPTZ	-
finished_at	TIMESTAMPTZ	-
error_summary	TEXT	-

raw_data_records [CORE]

Immutable-ish source payloads retained for audit/reprocessing.

PK: `raw_data_record_id`

FKs: `ingestion_batch_id -> ingestion_batches.ingestion_batch_id`

Column	Type	Declared flags / references
raw_data_record_id	UUID	PK; DEFAULT
ingestion_batch_id	UUID	FK -> ingestion_batches.ingestion_batch_id; NOT NULL
external_record_key	VARCHAR(240)	-
entity_type	VARCHAR(80)	NOT NULL
observed_at	TIMESTAMPTZ	-
received_at	TIMESTAMPTZ	NOT NULL; DEFAULT
payload	JSONB	NOT NULL
payload_hash	CHAR(64)	NOT NULL
validation_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK

Table-level constraints:

```
UNIQUE (ingestion_batch_id, payload_hash)
```

data_quality_rules [CORE]

Configurable checks and severity.

PK: `data_quality_rule_id`

FKs: `workspace_id -> workspaces.workspace_id`

Column	Type	Declared flags / references
data_quality_rule_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id
rule_key	VARCHAR(120)	NOT NULL
entity_type	VARCHAR(80)	NOT NULL
severity	VARCHAR(16)	NOT NULL; CHECK
rule_definition	JSONB	NOT NULL
is_enabled	BOOLEAN	NOT NULL; DEFAULT
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (workspace_id, rule_key)
```

data_quality_issues [CORE]

Detected missing intervals, duplicate, invalid or suspicious records.

PK: data_quality_issue_id

FKs: data_snapshot_id -> data_snapshots.data_snapshot_id; ingestion_batch_id -> ingestion_batches.ingestion_batch_id; data_quality_rule_id -> data_quality_rules.data_quality_rule_id; resolved_by_user_id -> users.user_id

Column	Type	Declared flags / references
data_quality_issue_id	UUID	PK; DEFAULT
data_snapshot_id	UUID	FK -> data_snapshots.data_snapshot_id
ingestion_batch_id	UUID	FK -> ingestion_batches.ingestion_batch_id
data_quality_rule_id	UUID	FK -> data_quality_rules.data_quality_rule_id
entity_type	VARCHAR(80)	NOT NULL
entity_id	UUID	-
field_name	VARCHAR(160)	-
severity	VARCHAR(16)	NOT NULL; CHECK
issue_code	VARCHAR(100)	NOT NULL
issue_detail	JSONB	NOT NULL; DEFAULT
issue_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
detected_at	TIMESTAMPTZ	NOT NULL; DEFAULT
resolved_at	TIMESTAMPTZ	-
resolved_by_user_id	UUID	FK -> users.user_id

refresh_policies [CORE]

Incremental refresh/backfill schedule per provider/category.

PK: refresh_policy_id

FKs: provider_connection_id -> provider_connections.provider_connection_id

Column	Type	Declared flags / references
refresh_policy_id	UUID	PK; DEFAULT
provider_connection_id	UUID	FK -> provider_connections.provider_connection_id; NOT NULL
data_category	VARCHAR(60)	NOT NULL
frequency	VARCHAR(40)	NOT NULL
schedule_expression	VARCHAR(120)	-
lookback_days	INTEGER	NOT NULL; DEFAULT; CHECK
backfill_enabled	BOOLEAN	NOT NULL; DEFAULT
is_enabled	BOOLEAN	NOT NULL; DEFAULT
last_started_at	TIMESTAMPTZ	-
last_success_at	TIMESTAMPTZ	-
next_run_at	TIMESTAMPTZ	-

Table-level constraints:

UNIQUE (provider_connection_id, data_category)

financial_metric_definitions [CORE]

Canonical fundamental metric vocabulary, units and derived formulas.

PK: financial_metric_definition_id

FKs: none declared

Column	Type	Declared flags / references
financial_metric_definition_id	UUID	PK; DEFAULT
metric_key	VARCHAR(120)	NOT NULL; UNIQUE
display_name	VARCHAR(160)	NOT NULL
description	TEXT	-
source_concept	VARCHAR(200)	-
canonical_unit	VARCHAR(40)	NOT NULL
value_type	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
calculation_definition	JSONB	NOT NULL; DEFAULT
is_derived	BOOLEAN	NOT NULL; DEFAULT

financial_statements [CORE]

Company statement/period/filing/source snapshot metadata.

PK: financial_statement_id

FKs: company_id -> companies.company_id; instrument_id -> instruments.instrument_id; source_registry_id -> source_registry.source_registry_id; data_snapshot_id -> data_snapshots.data_snapshot_id; reporting_currency -> currencies.currency_code

Column	Type	Declared flags / references
financial_statement_id	UUID	PK; DEFAULT
company_id	UUID	FK -> companies.company_id; NOT NULL
instrument_id	UUID	FK -> instruments.instrument_id
source_registry_id	UUID	FK -> source_registry.source_registry_id
data_snapshot_id	UUID	FK -> data_snapshots.data_snapshot_id
statement_type	VARCHAR(24)	NOT NULL; CHECK
period_type	VARCHAR(24)	NOT NULL; CHECK
fiscal_year	INTEGER	-
fiscal_period	VARCHAR(20)	-
period_start	DATE	-
period_end	DATE	NOT NULL
filed_at	TIMESTAMPTZ	-
accepted_at	TIMESTAMPTZ	-
accession_number	VARCHAR(80)	-
filing_url	TEXT	-
reporting_currency	CHAR(3)	FK -> currencies.currency_code
source_observed_at	TIMESTAMPTZ	-
recorded_at	TIMESTAMPTZ	NOT NULL; DEFAULT
superseded_at	TIMESTAMPTZ	-

Table-level constraints:

```
CHECK (period_start IS NULL OR period_end >= period_start)
UNIQUE (company_id, statement_type, period_type, period_end, accession_number)
```

financial_facts [CORE]

Normalized facts with dimensional context and bitemporal provenance.

PK: financial_fact_id

FKs: financial_statement_id -> financial_statements.financial_statement_id; financial_metric_definition_id -> financial_metric_definitions.financial_metric_definition_id; currency_code -> currencies.currency_code

Column	Type	Declared flags / references
financial_fact_id	UUID	PK; DEFAULT
financial_statement_id	UUID	FK -> financial_statements.financial_statement_id; NOT NULL
financial_metric_definition_id	UUID	FK -> financial_metric_definitions.financial_metric_definition_id; NOT NULL
reported_concept	VARCHAR(240)	-
numeric_value	NUMERIC(30,10)	-
text_value	TEXT	-
boolean_value	BOOLEAN	-
unit	VARCHAR(40)	NOT NULL
currency_code	CHAR(3)	FK -> currencies.currency_code
dimensions	JSONB	NOT NULL; DEFAULT
source_locator	VARCHAR(240)	-
fact_context	JSONB	NOT NULL; DEFAULT
valid_from	TIMESTAMPTZ	-
valid_to	TIMESTAMPTZ	-
recorded_at	TIMESTAMPTZ	NOT NULL; DEFAULT
superseded_at	TIMESTAMPTZ	-
content_hash	CHAR(64)	-

Table-level constraints:

```
CHECK (valid_to IS NULL OR valid_from IS NULL OR valid_to >= valid_from)
CHECK (num_nonnulls(numeric_value, text_value, boolean_value) <= 1)
```

price_bars [CORE]

OHLCV bars per instrument/timeframe/adjustment/snapshot with bitemporal metadata.

PK: price_bar_id

FKs: data_snapshot_id -> data_snapshots.data_snapshot_id; instrument_id -> instruments.instrument_id; currency_code -> currencies.currency_code

Column	Type	Declared flags / references
price_bar_id	UUID	PK; DEFAULT
data_snapshot_id	UUID	FK -> data_snapshots.data_snapshot_id; NOT NULL
instrument_id	UUID	FK -> instruments.instrument_id; NOT NULL
interval_code	VARCHAR(20)	NOT NULL; DEFAULT
bar_time	TIMESTAMPTZ	NOT NULL
open_price	NUMERIC(24,10)	NOT NULL; CHECK
high_price	NUMERIC(24,10)	NOT NULL; CHECK
low_price	NUMERIC(24,10)	NOT NULL; CHECK
close_price	NUMERIC(24,10)	NOT NULL; CHECK
volume	NUMERIC(30,6)	CHECK
currency_code	CHAR(3)	FK -> currencies.currency_code; NOT NULL
adjustment_type	VARCHAR(24)	NOT NULL; DEFAULT; CHECK
source_observed_at	TIMESTAMPTZ	-
valid_from	TIMESTAMPTZ	NOT NULL; DEFAULT
valid_to	TIMESTAMPTZ	-
recorded_at	TIMESTAMPTZ	NOT NULL; DEFAULT
superseded_at	TIMESTAMPTZ	-
quality_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK

Table-level constraints:

```
UNIQUE (data_snapshot_id, instrument_id, interval_code, bar_time, adjustment_type)
CHECK (high_price >= low_price)
CHECK (high_price >= GREATEST(open_price, close_price))
CHECK (low_price <= LEAST(open_price, close_price))
CHECK (valid_to IS NULL OR valid_to >= valid_from)
```

corporate_actions [CORE]

Dividends, splits and other dated corporate actions.

PK: corporate_action_id

FKs: data_snapshot_id -> data_snapshots.data_snapshot_id; instrument_id -> instruments.instrument_id; source_registry_id -> source_registry.source_registry_id; currency_code -> currencies.currency_code

Column	Type	Declared flags / references
corporate_action_id	UUID	PK; DEFAULT
data_snapshot_id	UUID	FK -> data_snapshots.data_snapshot_id; NOT NULL
instrument_id	UUID	FK -> instruments.instrument_id; NOT NULL
source_registry_id	UUID	FK -> source_registry.source_registry_id
action_type	VARCHAR(30)	NOT NULL; CHECK
announcement_at	TIMESTAMPTZ	-
ex_date	DATE	NOT NULL
record_date	DATE	-
payable_date	DATE	-
cash_amount	NUMERIC(24,10)	-
currency_code	CHAR(3)	FK -> currencies.currency_code
split_numerator	NUMERIC(20,8)	-
split_denominator	NUMERIC(20,8)	-
source_observed_at	TIMESTAMPTZ	-
recorded_at	TIMESTAMPTZ	NOT NULL; DEFAULT
action_details	JSONB	NOT NULL; DEFAULT

Table-level constraints:

```
CHECK (split_denominator IS NULL OR split_denominator <> 0)
UNIQUE (data_snapshot_id, instrument_id, action_type, ex_date)
```

index_constituents [CORE]

Time-aware index component membership and weights.

PK: index_constituent_id

FKs: index_instrument_id -> instruments.instrument_id; constituent_instrument_id -> instruments.instrument_id; data_snapshot_id -> data_snapshots.data_snapshot_id

Column	Type	Declared flags / references
index_constituent_id	UUID	PK; DEFAULT
index_instrument_id	UUID	FK -> instruments.instrument_id; NOT NULL
constituent_instrument_id	UUID	FK -> instruments.instrument_id; NOT NULL
data_snapshot_id	UUID	FK -> data_snapshots.data_snapshot_id
valid_from	DATE	NOT NULL
valid_to	DATE	-
weight	NUMERIC(18,10)	CHECK
recorded_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
CHECK (index_instrument_id <> constituent_instrument_id)
CHECK (valid_to IS NULL OR valid_to >= valid_from)
UNIQUE (index_instrument_id, constituent_instrument_id, valid_from)
```

news_items [CORE]

Attributed financial news items with timestamps/source metadata.

PK: news_item_id

FKs: source_registry_id -> source_registry.source_registry_id; data_snapshot_id -> data_snapshots.data_snapshot_id

Column	Type	Declared flags / references
news_item_id	UUID	PK; DEFAULT
source_registry_id	UUID	FK -> source_registry.source_registry_id
data_snapshot_id	UUID	FK -> data_snapshots.data_snapshot_id
headline	VARCHAR(600)	NOT NULL
summary	TEXT	-
article_url	TEXT	-
publisher_name	VARCHAR(200)	-
published_at	TIMESTAMPTZ	-
retrieved_at	TIMESTAMPTZ	NOT NULL; DEFAULT
language_code	VARCHAR(12)	-
content_hash	CHAR(64)	-
sentiment_score	NUMERIC(8,6)	-
metadata	JSONB	NOT NULL; DEFAULT

news_instruments [CORE]

News-to-instrument relevance mapping.

PK: news_item_id, instrument_id

FKs: news_item_id -> news_items.news_item_id; instrument_id -> instruments.instrument_id

Column	Type	Declared flags / references
news_item_id	UUID	PK; FK -> news_items.news_item_id
instrument_id	UUID	PK; FK -> instruments.instrument_id
relevance_score	NUMERIC(8,6)	-
linking_method	VARCHAR(30)	NOT NULL; DEFAULT

market_events [CORE]

Earnings, dividends, filings and other calendar events.

PK: market_event_id

FKs: instrument_id -> instruments.instrument_id; company_id -> companies.company_id; source_registry_id -> source_registry.source_registry_id; data_snapshot_id -> data_snapshots.data_snapshot_id

Column	Type	Declared flags / references
market_event_id	UUID	PK; DEFAULT
instrument_id	UUID	FK -> instruments.instrument_id
company_id	UUID	FK -> companies.company_id
source_registry_id	UUID	FK -> source_registry.source_registry_id
data_snapshot_id	UUID	FK -> data_snapshots.data_snapshot_id
event_type	VARCHAR(40)	NOT NULL; CHECK
event_title	VARCHAR(300)	NOT NULL
scheduled_at	TIMESTAMPTZ	-
event_date	DATE	-
event_status	VARCHAR(24)	NOT NULL; DEFAULT; CHECK
event_details	JSONB	NOT NULL; DEFAULT
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

technical_indicator_definitions [CORE]

Indicator formula/version and accepted parameters.

PK: technical_indicator_definition_id

FKs: none declared

Column	Type	Declared flags / references
technical_indicator_definition_id	UUID	PK; DEFAULT
indicator_key	VARCHAR(80)	NOT NULL; UNIQUE
display_name	VARCHAR(140)	NOT NULL
description	TEXT	-
calculation_version	VARCHAR(100)	NOT NULL
parameter_schema	JSONB	NOT NULL; DEFAULT
default_parameters	JSONB	NOT NULL; DEFAULT
output_units	VARCHAR(60)	-
is_enabled	BOOLEAN	NOT NULL; DEFAULT

indicator_profiles [CORE]

Named per-user default indicator sets.

PK: indicator_profile_id

FKs: workspace_id -> workspaces.workspace_id; owner_user_id -> users.user_id

Column	Type	Declared flags / references
indicator_profile_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id; NOT NULL
owner_user_id	UUID	FK -> users.user_id; NOT NULL
profile_name	VARCHAR(140)	NOT NULL
is_default	BOOLEAN	NOT NULL; DEFAULT
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

UNIQUE (workspace_id, owner_user_id, profile_name)

indicator_profile_items [CORE]

Configured indicators and parameters in a profile.

PK: `indicator_profile_id, technical_indicator_definition_id`

FKs: `indicator_profile_id -> indicator_profiles.indicator_profile_id; technical_indicator_definition_id -> technical_indicator_definitions.technical_indicator_definition_id`

Column	Type	Declared flags / references
indicator_profile_id	UUID	PK; FK -> indicator_profiles.indicator_profile_id
technical_indicator_definition_id	UUID	PK; FK -> technical_indicator_definitions.technical_indicator_definition_id
parameters	JSONB	NOT NULL; DEFAULT
display_order	INTEGER	NOT NULL; DEFAULT
enabled	BOOLEAN	NOT NULL; DEFAULT

3.4 Data explorer and lineage (6 tables)

Implementation scope
CORE TARGET SCHEMA - implement in staged releases according to dependencies and priorities.

structured_data_sources [CORE]

Approved external structured sources and read/write policy.

PK: `structured_data_source_id`

FKs: `workspace_id -> workspaces.workspace_id; owner_user_id -> users.user_id`

Column	Type	Declared flags / references
structured_data_source_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id; NOT NULL
owner_user_id	UUID	FK -> users.user_id; NOT NULL
source_name	VARCHAR(160)	NOT NULL
source_kind	VARCHAR(24)	NOT NULL; CHECK
connection_secret_ref	TEXT	-
non_secret_config	JSONB	NOT NULL; DEFAULT
allowlisted_objects	JSONB	NOT NULL; DEFAULT
access_mode	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
connection_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
last_tested_at	TIMESTAMPTZ	-
last_synced_at	TIMESTAMPTZ	-
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (workspace_id, source_name)
```

saved_queries [CORE]

Versionable saved SQL/visual/financial screen definitions.

PK: `saved_query_id`

FKs: `workspace_id -> workspaces.workspace_id; owner_user_id -> users.user_id; structured_data_source_id -> structured_data_sources.structured_data_source_id`

Column	Type	Declared flags / references
saved_query_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id; NOT NULL
owner_user_id	UUID	FK -> users.user_id; NOT NULL
structured_data_source_id	UUID	FK -> structured_data_sources.structured_data_source_id
query_name	VARCHAR(180)	NOT NULL
query_type	VARCHAR(24)	NOT NULL; CHECK
query_definition	JSONB	NOT NULL
parameter_schema	JSONB	NOT NULL; DEFAULT
read_only_required	BOOLEAN	NOT NULL; DEFAULT
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT
updated_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (workspace_id, owner_user_id, query_name)
```

user_datasets [CORE]

Imported and derived datasets, schema/storage pointer and snapshot link.

PK: user_dataset_id

FKs: workspace_id -> workspaces.workspace_id; owner_user_id -> users.user_id; parent_dataset_id -> user_datasets.user_dataset_id; data_snapshot_id -> data_snapshots.data_snapshot_id

Column	Type	Declared flags / references
user_dataset_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id; NOT NULL
owner_user_id	UUID	FK -> users.user_id; NOT NULL
parent_dataset_id	UUID	FK -> user_datasets.user_dataset_id
data_snapshot_id	UUID	FK -> data_snapshots.data_snapshot_id
dataset_name	VARCHAR(180)	NOT NULL
source_type	VARCHAR(24)	NOT NULL; CHECK
storage_uri	TEXT	-
schema_definition	JSONB	NOT NULL; DEFAULT
row_count	BIGINT	CHECK
content_hash	CHAR(64)	-
min_observation_date	DATE	-
max_observation_date	DATE	-
dataset_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (workspace_id, owner_user_id, dataset_name)
CHECK (max_observation_date IS NULL OR min_observation_date IS NULL OR max_observation_date >= min_observation_date)
```

dataset_transformations [CORE]

Ordered transformations and input/output lineage.

PK: dataset_transformation_id

FKs: workspace_id -> workspaces.workspace_id; input_dataset_id -> user_datasets.user_dataset_id; output_dataset_id -> user_datasets.user_dataset_id; execution_run_id -> execution_runs.execution_run_id; created_by_user_id -> users.user_id

Column	Type	Declared flags / references
dataset_transformation_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id; NOT NULL
input_dataset_id	UUID	FK -> user_datasets.user_dataset_id
output_dataset_id	UUID	FK -> user_datasets.user_dataset_id; NOT NULL
execution_run_id	UUID	FK -> execution_runs.execution_run_id
created_by_user_id	UUID	FK -> users.user_id
transformation_type	VARCHAR(60)	NOT NULL
transformation_order	INTEGER	NOT NULL; DEFAULT; CHECK
transformation_definition	JSONB	NOT NULL
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (output_dataset_id, transformation_order)
```

dataset_lineage [CORE]

Upstream dataset edges for reproducibility.

PK: user_dataset_id, upstream_dataset_id, lineage_role

FKs: user_dataset_id -> user_datasets.user_dataset_id; upstream_dataset_id -> user_datasets.user_dataset_id; dataset_transformation_id -> dataset_transformations.dataset_transformation_id

Column	Type	Declared flags / references
user_dataset_id	UUID	PK; FK -> user_datasets.user_dataset_id
upstream_dataset_id	UUID	PK; FK -> user_datasets.user_dataset_id
dataset_transformation_id	UUID	FK -> dataset_transformations.dataset_transformation_id
lineage_role	VARCHAR(40)	PK; DEFAULT

Table-level constraints:

```
CHECK (user_dataset_id <> upstream_dataset_id)
```

query_executions [CORE]

Validated query outcome, run ID, limits and output dataset.

PK: query_execution_id

FKs: saved_query_id -> saved_queries.saved_query_id; structured_data_source_id -> structured_data_sources.structured_data_source_id; execution_run_id -> execution_runs.execution_run_id; requested_by_user_id -> users.user_id; result_dataset_id -> user_datasets.user_dataset_id

Column	Type	Declared flags / references
query_execution_id	UUID	PK; DEFAULT
saved_query_id	UUID	FK -> saved_queries.saved_query_id
structured_data_source_id	UUID	FK -> structured_data_sources.structured_data_source_id
execution_run_id	UUID	FK -> execution_runs.execution_run_id
requested_by_user_id	UUID	FK -> users.user_id
query_text_redacted	TEXT	-
parameters	JSONB	NOT NULL; DEFAULT
validation_outcome	VARCHAR(20)	NOT NULL; CHECK
row_count	BIGINT	CHECK
result_dataset_id	UUID	FK -> user_datasets.user_dataset_id
executed_at	TIMESTAMPTZ	NOT NULL; DEFAULT
elapsed_ms	BIGINT	CHECK
error_message_redacted	TEXT	-

3.5 Research library and RAG (21 tables)

Implementation scope
CORE TARGET SCHEMA - implement in staged releases according to dependencies and priorities.

research_sources [CORE]

Document connector/source configuration.

PK: research_source_id

FKs: source_registry_id -> source_registry.source_registry_id; workspace_id -> workspaces.workspace_id

Column	Type	Declared flags / references
research_source_id	UUID	PK; DEFAULT
source_registry_id	UUID	FK -> source_registry.source_registry_id; NOT NULL
workspace_id	UUID	FK -> workspaces.workspace_id
source_name	VARCHAR(180)	NOT NULL
connector_kind	VARCHAR(40)	NOT NULL; CHECK
source_config	JSONB	NOT NULL; DEFAULT
sync_enabled	BOOLEAN	NOT NULL; DEFAULT
last_sync_at	TIMESTAMPTZ	-
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:
UNIQUE (workspace_id, source_name)

research_documents [CORE]

Canonical owned financial research document and searchable metadata.

PK: research_document_id

FKs: workspace_id -> workspaces.workspace_id; owner_user_id -> users.user_id; research_source_id -> research_sources.research_source_id; company_id -> companies.company_id; instrument_id -> instruments.instrument_id

Column	Type	Declared flags / references
research_document_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id; NOT NULL
owner_user_id	UUID	FK -> users.user_id; NOT NULL
research_source_id	UUID	FK -> research_sources.research_source_id
company_id	UUID	FK -> companies.company_id
instrument_id	UUID	FK -> instruments.instrument_id
document_type	VARCHAR(40)	NOT NULL; CHECK
title	VARCHAR(500)	NOT NULL
canonical_url	TEXT	-
publisher_name	VARCHAR(200)	-
language_code	VARCHAR(12)	-
published_at	TIMESTAMPTZ	-
retrieved_at	TIMESTAMPTZ	NOT NULL; DEFAULT
content_hash	CHAR(64)	-
lifecycle_status	VARCHAR(24)	NOT NULL; DEFAULT; CHECK
tags	TEXT[]	NOT NULL; DEFAULT
financial_metadata	JSONB	NOT NULL; DEFAULT
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT
updated_at	TIMESTAMPTZ	NOT NULL; DEFAULT

document_versions [CORE]

Immutable document versions, content hashes and extraction status.

PK: `document_version_id`

FKs: `research_document_id -> research_documents.research_document_id; supersedes_version_id -> document_versions.document_version_id`

Column	Type	Declared flags / references
document_version_id	UUID	PK; DEFAULT
research_document_id	UUID	FK -> research_documents.research_document_id; NOT NULL
version_number	INTEGER	NOT NULL; CHECK
supersedes_version_id	UUID	FK -> document_versions.document_version_id
mime_type	VARCHAR(120)	-
size_bytes	BIGINT	CHECK
malware_scan_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
content_hash	CHAR(64)	NOT NULL
original_storage_uri	TEXT	-
extracted_text_storage_uri	TEXT	-
extracted_text	TEXT	-
extraction_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
extracted_at	TIMESTAMPTZ	-
is_active	BOOLEAN	NOT NULL; DEFAULT
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (research_document_id, version_number)
UNIQUE (research_document_id, content_hash)
```

document_chunks [CORE]

Chunk text and exact page/section/character provenance.

PK: `document_chunk_id`

FKs: `document_version_id -> document_versions.document_version_id`

Column	Type	Declared flags / references
document_chunk_id	UUID	PK; DEFAULT
document_version_id	UUID	FK -> document_versions.document_version_id; NOT NULL
chunk_index	INTEGER	NOT NULL; CHECK
text_content	TEXT	NOT NULL
token_count	INTEGER	CHECK
page_number	INTEGER	CHECK
section_path	TEXT	-
character_start	BIGINT	-
character_end	BIGINT	-
paragraph_start	INTEGER	-
paragraph_end	INTEGER	-
table_index	INTEGER	-
metadata	JSONB	NOT NULL; DEFAULT
content_hash	CHAR(64)	NOT NULL
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (document_version_id, chunk_index)
CHECK (character_end IS NULL OR character_start IS NULL OR character_end >= character_start)
```

document_tables [CORE]

Extracted financial tables and their header structure.

PK: document_table_id

FKs: document_version_id -> document_versions.document_version_id

Column	Type	Declared flags / references
document_table_id	UUID	PK; DEFAULT
document_version_id	UUID	FK -> document_versions.document_version_id; NOT NULL
table_index	INTEGER	NOT NULL; CHECK
page_number	INTEGER	-
caption	TEXT	-
header_rows	JSONB	NOT NULL; DEFAULT
table_metadata	JSONB	NOT NULL; DEFAULT
extraction_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK

Table-level constraints:

```
UNIQUE (document_version_id, table_index)
```

document_table_cells [CORE]

Row/column-level table cells, units and reporting-period labels.

PK: document_table_cell_id

FKs: document_table_id -> document_tables.document_table_id; currency_code -> currencies.currency_code

Column	Type	Declared flags / references
document_table_cell_id	UUID	PK; DEFAULT
document_table_id	UUID	FK -> document_tables.document_table_id; NOT NULL
row_index	INTEGER	NOT NULL; CHECK
column_index	INTEGER	NOT NULL; CHECK
cell_text	TEXT	-
row_header	TEXT	-
column_header	TEXT	-
numeric_value	NUMERIC(30,10)	-
unit	VARCHAR(40)	-
currency_code	CHAR(3)	FK -> currencies.currency_code
reporting_period_label	VARCHAR(80)	-
cell_metadata	JSONB	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (document_table_id, row_index, column_index)
```

document_access_grants [CORE]

Explicit user/workspace access to private research docs.

PK: document_access_grant_id

FKs: research_document_id -> research_documents.research_document_id; grantee_user_id -> users.user_id; grantee_workspace_id -> workspaces.workspace_id; granted_by_user_id -> users.user_id

Column	Type	Declared flags / references
document_access_grant_id	UUID	PK; DEFAULT
research_document_id	UUID	FK -> research_documents.research_document_id; NOT NULL
grantee_user_id	UUID	FK -> users.user_id
grantee_workspace_id	UUID	FK -> workspaces.workspace_id
permission_level	VARCHAR(20)	NOT NULL; CHECK
granted_by_user_id	UUID	FK -> users.user_id; NOT NULL
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
CHECK (num_nonnulls(grantee_user_id, grantee_workspace_id) = 1)
```

knowledge_indexes [CORE]

Embedding model/dimension/chunk-policy version for one vector index.

PK: knowledge_index_id

FKs: workspace_id -> workspaces.workspace_id; ai_model_id -> ai_models.ai_model_id

Column	Type	Declared flags / references
knowledge_index_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id
index_name	VARCHAR(180)	NOT NULL
ai_model_id	UUID	FK -> ai_models.ai_model_id; NOT NULL
embedding_dimension	INTEGER	NOT NULL; CHECK
distance_metric	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
chunking_policy	JSONB	NOT NULL; DEFAULT
index_version	INTEGER	NOT NULL; DEFAULT; CHECK
index_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (workspace_id, index_name, index_version)
```

chunk_embeddings [CORE]

pgvector embedding keyed to stable chunk and index IDs.

PK: chunk_embedding_id

FKs: document_chunk_id -> document_chunks.document_chunk_id; knowledge_index_id -> knowledge_indexes.knowledge_index_id

Column	Type	Declared flags / references
chunk_embedding_id	UUID	PK; DEFAULT
document_chunk_id	UUID	FK -> document_chunks.document_chunk_id; NOT NULL
knowledge_index_id	UUID	FK -> knowledge_indexes.knowledge_index_id; NOT NULL
embedding	VECTOR	NOT NULL
embedding_dimension	INTEGER	NOT NULL; CHECK
embedding_model_key	VARCHAR(160)	NOT NULL
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (document_chunk_id, knowledge_index_id)
CHECK (vector_dims(embedding) = embedding_dimension)
```

document_index_jobs [CORE]

Index/reindex/delete job state per document version and index.

PK: document_index_job_id

FKs: document_version_id -> document_versions.document_version_id; knowledge_index_id -> knowledge_indexes.knowledge_index_id; background_job_id -> background_jobs.background_job_id

Column	Type	Declared flags / references
document_index_job_id	UUID	PK; DEFAULT
document_version_id	UUID	FK -> document_versions.document_version_id; NOT NULL
knowledge_index_id	UUID	FK -> knowledge_indexes.knowledge_index_id; NOT NULL
background_job_id	UUID	FK -> background_jobs.background_job_id
status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
operation_key	VARCHAR(200)	NOT NULL; UNIQUE
chunks_expected	INTEGER	-
chunks_indexed	INTEGER	NOT NULL; DEFAULT; CHECK
started_at	TIMESTAMPTZ	-
finished_at	TIMESTAMPTZ	-
last_error_redacted	TEXT	-

Table-level constraints:

```
UNIQUE (document_version_id, knowledge_index_id, operation_key)
```

index_reconciliation_events [CORE]

Retryable reconciliation for failed/stale vector rows and deletions.

PK: index_reconciliation_event_id

FKs: document_version_id -> document_versions.document_version_id; knowledge_index_id -> knowledge_indexes.knowledge_index_id; document_chunk_id -> document_chunks.document_chunk_id; background_job_id -> background_jobs.background_job_id

Column	Type	Declared flags / references
index_reconciliation_event_id	UUID	PK; DEFAULT
document_version_id	UUID	FK -> document_versions.document_version_id
knowledge_index_id	UUID	FK -> knowledge_indexes.knowledge_index_id
document_chunk_id	UUID	FK -> document_chunks.document_chunk_id
background_job_id	UUID	FK -> background_jobs.background_job_id
operation_type	VARCHAR(30)	NOT NULL; CHECK
outcome	VARCHAR(24)	NOT NULL; CHECK
retry_count	SMALLINT	NOT NULL; DEFAULT; CHECK
detail_redacted	JSONB	NOT NULL; DEFAULT
occurred_at	TIMESTAMPTZ	NOT NULL; DEFAULT

research_collections [CORE]

User/workspace research collections.

PK: research_collection_id

FKs: workspace_id -> workspaces.workspace_id; owner_user_id -> users.user_id

Column	Type	Declared flags / references
research_collection_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id; NOT NULL
owner_user_id	UUID	FK -> users.user_id; NOT NULL
collection_name	VARCHAR(180)	NOT NULL
description	TEXT	-
collection_type	VARCHAR(40)	NOT NULL; DEFAULT
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT
updated_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (workspace_id, owner_user_id, collection_name)
```

research_collection_documents [CORE]

Collection membership for research documents.

PK: research_collection_id, research_document_id

FKs: research_collection_id -> research_collections.research_collection_id; research_document_id -> research_documents.research_document_id; added_by_user_id -> users.user_id

Column	Type	Declared flags / references
research_collection_id	UUID	PK; FK -> research_collections.research_collection_id
research_document_id	UUID	PK; FK -> research_documents.research_document_id
added_by_user_id	UUID	FK -> users.user_id
added_at	TIMESTAMPTZ	NOT NULL; DEFAULT

research_collection_conversations [CORE]

Conversation membership in a saved research collection.

PK: research_collection_id, conversation_id

FKs: research_collection_id -> research_collections.research_collection_id; conversation_id -> conversations.conversation_id; added_by_user_id -> users.user_id

Column	Type	Declared flags / references
research_collection_id	UUID	PK; FK -> research_collections.research_collection_id
conversation_id	UUID	PK; FK -> conversations.conversation_id
added_by_user_id	UUID	FK -> users.user_id
added_at	TIMESTAMPTZ	NOT NULL; DEFAULT

research_collection_reports [CORE]

Saved-report membership in a research collection.

PK: research_collection_id, report_id

FKs: research_collection_id -> research_collections.research_collection_id; report_id -> reports.report_id; added_by_user_id -> users.user_id

Column	Type	Declared flags / references
research_collection_id	UUID	PK; FK -> research_collections.research_collection_id
report_id	UUID	PK; FK -> reports.report_id
added_by_user_id	UUID	FK -> users.user_id
added_at	TIMESTAMPTZ	NOT NULL; DEFAULT

conversations [CORE]

Owned AI-assistant conversations and selected RAG/model context.

PK: conversation_id

FKs: workspace_id -> workspaces.workspace_id; owner_user_id -> users.user_id; selected_ai_model_id -> ai_models.ai_model_id

Column	Type	Declared flags / references
conversation_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id; NOT NULL
owner_user_id	UUID	FK -> users.user_id; NOT NULL
title	VARCHAR(300)	NOT NULL; DEFAULT
conversation_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
rag_enabled	BOOLEAN	NOT NULL; DEFAULT
selected_ai_model_id	UUID	FK -> ai_models.ai_model_id
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT
updated_at	TIMESTAMPTZ	NOT NULL; DEFAULT
deleted_at	TIMESTAMPTZ	-

messages [CORE]

Conversation messages, separate generations and inference config.

PK: message_id

FKs: conversation_id -> conversations.conversation_id; parent_message_id -> messages.message_id; ai_model_id -> ai_models.ai_model_id; ai_config_version_id -> ai_config_versions.ai_config_version_id

Column	Type	Declared flags / references
message_id	UUID	PK; DEFAULT
conversation_id	UUID	FK -> conversations.conversation_id; NOT NULL
parent_message_id	UUID	FK -> messages.message_id
role	VARCHAR(20)	NOT NULL; CHECK
content	TEXT	-
structured_content	JSONB	NOT NULL; DEFAULT
message_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
generation_number	INTEGER	NOT NULL; DEFAULT; CHECK
ai_model_id	UUID	FK -> ai_models.ai_model_id
ai_config_version_id	UUID	FK -> ai_config_versions.ai_config_version_id
prompt_tokens	INTEGER	CHECK
completion_tokens	INTEGER	CHECK
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT
completed_at	TIMESTAMPTZ	-

Table-level constraints:

UNIQUE (parent_message_id, generation_number)

message_attachments [CORE]

Validated file metadata and external object-store pointer.

PK: `message_attachment_id`

FKs: `message_id -> messages.message_id; uploaded_by_user_id -> users.user_id`

Column	Type	Declared flags / references
message_attachment_id	UUID	PK; DEFAULT
message_id	UUID	FK -> messages.message_id; NOT NULL
uploaded_by_user_id	UUID	FK -> users.user_id; NOT NULL
original_filename	VARCHAR(500)	NOT NULL
mime_type	VARCHAR(120)	NOT NULL
size_bytes	BIGINT	NOT NULL; CHECK
content_hash	CHAR(64)	-
storage_uri	TEXT	NOT NULL
processing_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

message_feedback [CORE]

Feedback on a specific AI response.

PK: `message_feedback_id`

FKs: `message_id -> messages.message_id; user_id -> users.user_id`

Column	Type	Declared flags / references
message_feedback_id	UUID	PK; DEFAULT
message_id	UUID	FK -> messages.message_id; NOT NULL
user_id	UUID	FK -> users.user_id; NOT NULL
rating	VARCHAR(20)	NOT NULL; CHECK
comment	TEXT	-
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (message_id, user_id)
```

retrieval_runs [CORE]

Persisted RAG query/config/index/filter trace.

PK: `retrieval_run_id`

FKs: `conversation_id -> conversations.conversation_id; message_id -> messages.message_id; workspace_id -> workspaces.workspace_id; requested_by_user_id -> users.user_id; rag_config_version_id -> rag_config_versions.rag_config_version_id; knowledge_index_id -> knowledge_indexes.knowledge_index_id; query_embedding_model_id -> ai_models.ai_model_id`

Column	Type	Declared flags / references
retrieval_run_id	UUID	PK; DEFAULT
conversation_id	UUID	FK -> conversations.conversation_id
message_id	UUID	FK -> messages.message_id
workspace_id	UUID	FK -> workspaces.workspace_id; NOT NULL
requested_by_user_id	UUID	FK -> users.user_id
rag_config_version_id	UUID	FK -> rag_config_versions.rag_config_version_id
knowledge_index_id	UUID	FK -> knowledge_indexes.knowledge_index_id
query_text_redacted	TEXT	-
query_embedding_model_id	UUID	FK -> ai_models.ai_model_id
filter_metadata	JSONB	NOT NULL; DEFAULT
status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
candidates_count	INTEGER	NOT NULL; DEFAULT; CHECK
returned_count	INTEGER	NOT NULL; DEFAULT; CHECK
permission_filtered_count	INTEGER	NOT NULL; DEFAULT; CHECK
elapsed_ms	BIGINT	CHECK
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

retrieval_run_chunks [CORE]

Ranked chunk results and whether permitted context was included.

PK: `retrieval_run_id, document_chunk_id`

FKs: `retrieval_run_id -> retrieval_runs.retrieval_run_id; document_chunk_id -> document_chunks.document_chunk_id`

Column	Type	Declared flags / references
retrieval_run_id	UUID	PK; FK -> retrieval_runs.retrieval_run_id
document_chunk_id	UUID	PK; FK -> document_chunks.document_chunk_id
rank_position	INTEGER	NOT NULL; CHECK
similarity_score	NUMERIC(12,10)	-
included_in_context	BOOLEAN	NOT NULL; DEFAULT
exclusion_reason	VARCHAR(100)	-
source_page	INTEGER	-
source_section	TEXT	-

Table-level constraints:

```
UNIQUE (retrieval_run_id, rank_position)
```

3.6 Portfolio and dashboard analytics (12 tables)

Implementation scope
CORE TARGET SCHEMA - implement in staged releases according to dependencies and priorities.

portfolios [CORE]

Manual/imported/paper portfolio container, currency and benchmark.

PK: `portfolio_id`

FKs: `workspace_id -> workspaces.workspace_id; owner_user_id -> users.user_id; base_currency -> currencies.currency_code; benchmark_instrument_id -> instruments.instrument_id`

Column	Type	Declared flags / references
portfolio_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id; NOT NULL
owner_user_id	UUID	FK -> users.user_id; NOT NULL
portfolio_name	VARCHAR(180)	NOT NULL
portfolio_type	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
base_currency	CHAR(3)	FK -> currencies.currency_code; NOT NULL
benchmark_instrument_id	UUID	FK -> instruments.instrument_id
data_source_label	VARCHAR(160)	-
portfolio_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
settings	JSONB	NOT NULL; DEFAULT
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT
updated_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (workspace_id, owner_user_id, portfolio_name)
```

portfolio_transactions [CORE]

Portfolio activity ledger, cash flows, trades, fees and corporate action entries.

PK: `portfolio_transaction_id`

FKs: `portfolio_id -> portfolios.portfolio_id; instrument_id -> instruments.instrument_id; data_snapshot_id -> data_snapshots.data_snapshot_id; currency_code -> currencies.currency_code; created_by_user_id -> users.user_id`

Column	Type	Declared flags / references
portfolio_transaction_id	UUID	PK; DEFAULT
portfolio_id	UUID	FK -> portfolios.portfolio_id; NOT NULL
instrument_id	UUID	FK -> instruments.instrument_id
data_snapshot_id	UUID	FK -> data_snapshots.data_snapshot_id
transaction_type	VARCHAR(24)	NOT NULL; CHECK
transaction_time	TIMESTAMPTZ	NOT NULL
settlement_date	DATE	-
quantity	NUMERIC(28,10)	-
unit_price	NUMERIC(24,10)	-
gross_amount	NUMERIC(28,10)	-
fees	NUMERIC(24,10)	NOT NULL; DEFAULT; CHECK
currency_code	CHAR(3)	FK -> currencies.currency_code; NOT NULL
external_transaction_id	VARCHAR(200)	-
notes	TEXT	-
created_by_user_id	UUID	FK -> users.user_id
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (portfolio_id, external_transaction_id)
```

portfolio_snapshots [CORE]

Point-in-time portfolio valuation with data snapshot and limitations.

PK: `portfolio_snapshot_id`

FKs: `portfolio_id -> portfolios.portfolio_id; data_snapshot_id -> data_snapshots.data_snapshot_id; base_currency -> currencies.currency_code`

Column	Type	Declared flags / references
portfolio_snapshot_id	UUID	PK; DEFAULT
portfolio_id	UUID	FK -> portfolios.portfolio_id; NOT NULL
data_snapshot_id	UUID	FK -> data_snapshots.data_snapshot_id
as_of_at	TIMESTAMPTZ	NOT NULL
total_market_value	NUMERIC(30,10)	-
cash_value	NUMERIC(30,10)	-
total_equity_value	NUMERIC(30,10)	-
realized_pnl	NUMERIC(30,10)	-
unrealized_pnl	NUMERIC(30,10)	-
base_currency	CHAR(3)	FK -> currencies.currency_code; NOT NULL
valuation_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
limitation_detail	JSONB	NOT NULL; DEFAULT
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (portfolio_id, as_of_at, data_snapshot_id)
```

portfolio_positions [CORE]

Holdings/weights/cost/P&L at a specific portfolio snapshot.

PK: `portfolio_position_id`

FKs: `portfolio_snapshot_id -> portfolio_snapshots.portfolio_snapshot_id; instrument_id -> instruments.instrument_id; currency_code -> currencies.currency_code`

Column	Type	Declared flags / references
portfolio_position_id	UUID	PK; DEFAULT
portfolio_snapshot_id	UUID	FK -> portfolio_snapshots.portfolio_snapshot_id; NOT NULL
instrument_id	UUID	FK -> instruments.instrument_id; NOT NULL
quantity	NUMERIC(28,10)	NOT NULL
average_cost	NUMERIC(24,10)	-
market_price	NUMERIC(24,10)	-
market_value	NUMERIC(30,10)	-
cost_basis	NUMERIC(30,10)	-
realized_pnl	NUMERIC(30,10)	-
unrealized_pnl	NUMERIC(30,10)	-
portfolio_weight	NUMERIC(18,10)	-
currency_code	CHAR(3)	FK -> currencies.currency_code
valuation_source	VARCHAR(30)	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (portfolio_snapshot_id, instrument_id)
```

portfolio_cash_balances [CORE]

Cash by currency at portfolio snapshot.

PK: `portfolio_snapshot_id, currency_code`

FKs: `portfolio_snapshot_id -> portfolio_snapshots.portfolio_snapshot_id; currency_code -> currencies.currency_code`

Column	Type	Declared flags / references
portfolio_snapshot_id	UUID	PK; FK -> portfolio_snapshots.portfolio_snapshot_id
currency_code	CHAR(3)	PK; FK -> currencies.currency_code
cash_balance	NUMERIC(30,10)	NOT NULL

portfolio_returns [CORE]

Daily/period portfolio and benchmark returns.

PK: portfolio_return_id

FKs: portfolio_id -> portfolios.portfolio_id; data_snapshot_id -> data_snapshots.data_snapshot_id

Column	Type	Declared flags / references
portfolio_return_id	UUID	PK; DEFAULT
portfolio_id	UUID	FK -> portfolios.portfolio_id; NOT NULL
data_snapshot_id	UUID	FK -> data_snapshots.data_snapshot_id
observation_date	DATE	NOT NULL
return_frequency	VARCHAR(20)	NOT NULL; DEFAULT
period_return	NUMERIC(18,10)	-
cumulative_return	NUMERIC(18,10)	-
benchmark_return	NUMERIC(18,10)	-
portfolio_value	NUMERIC(30,10)	-
is_partial	BOOLEAN	NOT NULL; DEFAULT
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

UNIQUE (portfolio_id, observation_date, return_frequency, data_snapshot_id)

watchlists [CORE]

Named user watchlists.

PK: watchlist_id

FKs: workspace_id -> workspaces.workspace_id; owner_user_id -> users.user_id

Column	Type	Declared flags / references
watchlist_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id; NOT NULL
owner_user_id	UUID	FK -> users.user_id; NOT NULL
watchlist_name	VARCHAR(160)	NOT NULL
display_order	INTEGER	NOT NULL; DEFAULT
display_preferences	JSONB	NOT NULL; DEFAULT
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT
updated_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

UNIQUE (workspace_id, owner_user_id, watchlist_name)

watchlist_items [CORE]

Instrument membership and ordering.

PK: watchlist_id, instrument_id

FKs: watchlist_id -> watchlists.watchlist_id; instrument_id -> instruments.instrument_id

Column	Type	Declared flags / references
watchlist_id	UUID	PK; FK -> watchlists.watchlist_id
instrument_id	UUID	PK; FK -> instruments.instrument_id
display_order	INTEGER	NOT NULL; DEFAULT
added_at	TIMESTAMPTZ	NOT NULL; DEFAULT
note	TEXT	-

dashboard_layouts [CORE]

Saved dashboard layouts/preferences.

PK: dashboard_layout_id

FKs: workspace_id -> workspaces.workspace_id; owner_user_id -> users.user_id

Column	Type	Declared flags / references
dashboard_layout_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id; NOT NULL
owner_user_id	UUID	FK -> users.user_id; NOT NULL
layout_name	VARCHAR(160)	NOT NULL
is_default	BOOLEAN	NOT NULL; DEFAULT
layout_version	INTEGER	NOT NULL; DEFAULT; CHECK
layout_preferences	JSONB	NOT NULL; DEFAULT
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT
updated_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

UNIQUE (workspace_id, owner_user_id, layout_name)

dashboard_widgets [CORE]

Widget positions and configuration.

PK: dashboard_widget_id

FKs: dashboard_layout_id -> dashboard_layouts.dashboard_layout_id

Column	Type	Declared flags / references
dashboard_widget_id	UUID	PK; DEFAULT
dashboard_layout_id	UUID	FK -> dashboard_layouts.dashboard_layout_id; NOT NULL
widget_key	VARCHAR(120)	NOT NULL
title	VARCHAR(180)	-
position_x	INTEGER	NOT NULL; DEFAULT
position_y	INTEGER	NOT NULL; DEFAULT
width_units	INTEGER	NOT NULL; DEFAULT; CHECK
height_units	INTEGER	NOT NULL; DEFAULT; CHECK
widget_config	JSONB	NOT NULL; DEFAULT
display_order	INTEGER	NOT NULL; DEFAULT

alert_rules [CORE]

Validated price/indicator/screen/signal/workflow alert definitions.

PK: alert_rule_id

FKs: workspace_id -> workspaces.workspace_id; owner_user_id -> users.user_id; instrument_id -> instruments.instrument_id; watchlist_id -> watchlists.watchlist_id

Column	Type	Declared flags / references
alert_rule_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id; NOT NULL
owner_user_id	UUID	FK -> users.user_id; NOT NULL
alert_name	VARCHAR(180)	NOT NULL
alert_type	VARCHAR(40)	NOT NULL; CHECK
rule_definition	JSONB	NOT NULL
instrument_id	UUID	FK -> instruments.instrument_id
watchlist_id	UUID	FK -> watchlists.watchlist_id
is_enabled	BOOLEAN	NOT NULL; DEFAULT
cooldown_seconds	INTEGER	NOT NULL; DEFAULT; CHECK
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT
updated_at	TIMESTAMPTZ	NOT NULL; DEFAULT

alert_events [CORE]

Deduplicated rule-triggered events with observations and timestamps.

PK: alert_event_id

FKs: alert_rule_id -> alert_rules.alert_rule_id; instrument_id -> instruments.instrument_id; data_snapshot_id -> data_snapshots.data_snapshot_id; acknowledged_by_user_id -> users.user_id

Column	Type	Declared flags / references
alert_event_id	UUID	PK; DEFAULT
alert_rule_id	UUID	FK -> alert_rules.alert_rule_id; NOT NULL
instrument_id	UUID	FK -> instruments.instrument_id
data_snapshot_id	UUID	FK -> data_snapshots.data_snapshot_id
deduplication_key	VARCHAR(240)	NOT NULL; UNIQUE
triggering_observation	JSONB	NOT NULL
trigger_condition	JSONB	NOT NULL
event_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
triggered_at	TIMESTAMPTZ	NOT NULL; DEFAULT
acknowledged_at	TIMESTAMPTZ	-
acknowledged_by_user_id	UUID	FK -> users.user_id

3.7 DCF valuation and quantitative risk (22 tables)

Implementation scope
CORE TARGET SCHEMA - implement in staged releases according to dependencies and priorities.

valuation_model_versions [CORE]

Versioned DCF formulas and implementation hash.

PK: `valuation_model_version_id`
FKs: `created_by_user_id -> users.user_id`

Column	Type	Declared flags / references
valuation_model_version_id	UUID	PK; DEFAULT
model_key	VARCHAR(80)	NOT NULL
version_number	INTEGER	NOT NULL; CHECK
methodology	VARCHAR(40)	NOT NULL; DEFAULT; CHECK
formula_version	VARCHAR(120)	NOT NULL
implementation_version	VARCHAR(120)	NOT NULL
implementation_hash	CHAR(64)	-
model_config	JSONB	NOT NULL; DEFAULT
created_by_user_id	UUID	FK -> users.user_id
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:
`UNIQUE (model_key, version_number)`

valuation_runs [CORE]

Reproducible company DCF run and headline outputs tied to data/model versions.

PK: valuation_run_id

FKs: execution_run_id -> execution_runs.execution_run_id; workspace_id -> workspaces.workspace_id; owner_user_id -> users.user_id; company_id -> companies.company_id; instrument_id -> instruments.instrument_id; valuation_model_version_id -> valuation_model_versions.valuation_model_version_id; data_snapshot_id -> data_snapshots.data_snapshot_id; parent_valuation_run_id -> valuation_runs.valuation_run_id; output_currency -> currencies.currency_code

Column	Type	Declared flags / references
valuation_run_id	UUID	PK; DEFAULT
execution_run_id	UUID	FK -> execution_runs.execution_run_id; NOT NULL; UNIQUE
workspace_id	UUID	FK -> workspaces.workspace_id; NOT NULL
owner_user_id	UUID	FK -> users.user_id; NOT NULL
company_id	UUID	FK -> companies.company_id; NOT NULL
instrument_id	UUID	FK -> instruments.instrument_id
valuation_model_version_id	UUID	FK -> valuation_model_versions.valuation_model_version_id; NOT NULL
data_snapshot_id	UUID	FK -> data_snapshots.data_snapshot_id; NOT NULL
parent_valuation_run_id	UUID	FK -> valuation_runs.valuation_run_id
scenario_name	VARCHAR(160)	NOT NULL; DEFAULT
valuation_date	DATE	NOT NULL
financial_period_end	DATE	-
output_currency	CHAR(3)	FK -> currencies.currency_code; NOT NULL
forecast_years	SMALLINT	NOT NULL; DEFAULT; CHECK
wacc	NUMERIC(14,10)	-
terminal_growth_rate	NUMERIC(14,10)	-
terminal_value	NUMERIC(30,10)	-
present_value_forecast_cash_flows	NUMERIC(30,10)	-
present_value_terminal_value	NUMERIC(30,10)	-
enterprise_value	NUMERIC(30,10)	-
cash_and_equivalents	NUMERIC(30,10)	-
total_debt	NUMERIC(30,10)	-
equity_value	NUMERIC(30,10)	-
diluted_shares_outstanding	NUMERIC(30,10)	-
value_per_share	NUMERIC(24,10)	-
valuation_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
data_quality_warnings	JSONB	NOT NULL; DEFAULT
limitations	JSONB	NOT NULL; DEFAULT
input_manifest	JSONB	NOT NULL; DEFAULT
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
CHECK (wacc IS NULL OR (wacc > -1 AND wacc < 1))
CHECK (terminal_growth_rate IS NULL OR (terminal_growth_rate > -1 AND terminal_growth_rate < 1))
CHECK (wacc IS NULL OR terminal_growth_rate IS NULL OR terminal_growth_rate < wacc)
```

valuation_assumptions [CORE]

Assumption value, year, unit, formula and fact/document/user provenance.

PK: valuation_assumption_id

FKs: valuation_run_id -> valuation_runs.valuation_run_id; source_financial_fact_id -> financial_facts.financial_fact_id; source_document_chunk_id -> document_chunks.document_chunk_id

Column	Type	Declared flags / references
valuation_assumption_id	UUID	PK; DEFAULT
valuation_run_id	UUID	FK -> valuation_runs.valuation_run_id; NOT NULL
assumption_key	VARCHAR(100)	NOT NULL
forecast_year	SMALLINT	-
numeric_value	NUMERIC(30,12)	-
text_value	TEXT	-
unit	VARCHAR(40)	NOT NULL
source_type	VARCHAR(24)	NOT NULL; CHECK
source_financial_fact_id	UUID	FK -> financial_facts.financial_fact_id
source_document_chunk_id	UUID	FK -> document_chunks.document_chunk_id
source_url	TEXT	-
formula_or_method	TEXT	-
is_user_approved	BOOLEAN	NOT NULL; DEFAULT
provenance_detail	JSONB	NOT NULL; DEFAULT
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (valuation_run_id, assumption_key, forecast_year)
CHECK (source_type <> 'source_fact' OR source_financial_fact_id IS NOT NULL)
CHECK (source_type <> 'document_evidence' OR source_document_chunk_id IS NOT NULL)
```

valuation_forecasts [CORE]

Annual revenue, margin, tax, NOPAT, capex, working capital, FCF and PV outputs.

PK: valuation_forecast_id

FKs: valuation_run_id -> valuation_runs.valuation_run_id

Column	Type	Declared flags / references
valuation_forecast_id	UUID	PK; DEFAULT
valuation_run_id	UUID	FK -> valuation_runs.valuation_run_id; NOT NULL
forecast_year	SMALLINT	NOT NULL; CHECK
fiscal_year	INTEGER	-
revenue	NUMERIC(30,10)	-
revenue_growth_rate	NUMERIC(14,10)	-
operating_income	NUMERIC(30,10)	-
operating_margin	NUMERIC(14,10)	-
tax_rate	NUMERIC(14,10)	-
nopat	NUMERIC(30,10)	-
depreciation_amortization	NUMERIC(30,10)	-
capital_expenditure	NUMERIC(30,10)	-
change_in_working_capital	NUMERIC(30,10)	-
free_cash_flow	NUMERIC(30,10)	-
discount_factor	NUMERIC(24,14)	-
present_value_fcf	NUMERIC(30,10)	-
calculation_detail	JSONB	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (valuation_run_id, forecast_year)
```

valuation_sensitivity_runs [CORE]

WACC/terminal-growth or other sensitivity grid definition.

PK: valuation_sensitivity_run_id

FKs: base_valuation_run_id -> valuation_runs.valuation_run_id

Column	Type	Declared flags / references
valuation_sensitivity_run_id	UUID	PK; DEFAULT
base_valuation_run_id	UUID	FK -> valuation_runs.valuation_run_id; NOT NULL
variable_x	VARCHAR(100)	NOT NULL; DEFAULT
variable_y	VARCHAR(100)	NOT NULL; DEFAULT
x_values	NUMERIC(14,10)[]	NOT NULL
y_values	NUMERIC(14,10)[]	NOT NULL
status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

valuation_sensitivity_points [CORE]

Computed enterprise/equity/per-share output at each sensitivity point.

PK: valuation_sensitivity_run_id, x_value, y_value

FKs: valuation_sensitivity_run_id -> valuation_sensitivity_runs.valuation_sensitivity_run_id

Column	Type	Declared flags / references
valuation_sensitivity_run_id	UUID	PK; FK -> valuation_sensitivity_runs.valuation_sensitivity_run_id
x_value	NUMERIC(14,10)	PK
y_value	NUMERIC(14,10)	PK
enterprise_value	NUMERIC(30,10)	-
equity_value	NUMERIC(30,10)	-
value_per_share	NUMERIC(24,10)	-
result_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
limitation	TEXT	-

valuation_run_comparisons [CORE]

Named groups for comparing saved valuation outputs.

PK: valuation_run_comparison_id

FKs: workspace_id -> workspaces.workspace_id; owner_user_id -> users.user_id

Column	Type	Declared flags / references
valuation_run_comparison_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id; NOT NULL
owner_user_id	UUID	FK -> users.user_id; NOT NULL
comparison_name	VARCHAR(180)	NOT NULL
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

UNIQUE (workspace_id, owner_user_id, comparison_name)

valuation_comparison_members [CORE]

Valuation runs included in a saved comparison.

PK: valuation_run_comparison_id, valuation_run_id

FKs: valuation_run_comparison_id -> valuation_run_comparisons.valuation_run_comparison_id; valuation_run_id -> valuation_runs.valuation_run_id

Column	Type	Declared flags / references
valuation_run_comparison_id	UUID	PK; FK -> valuation_run_comparisons.valuation_run_comparison_id
valuation_run_id	UUID	PK; FK -> valuation_runs.valuation_run_id
display_order	INTEGER	NOT NULL; DEFAULT
comparison_label	VARCHAR(160)	-

report_data_snapshots [CORE]

FK-backed report-to-input-snapshot lineage.

PK: report_id, data_snapshot_id

FKs: report_id -> reports.report_id; data_snapshot_id -> data_snapshots.data_snapshot_id

Column	Type	Declared flags / references
report_id	UUID	PK; FK -> reports.report_id
data_snapshot_id	UUID	PK; FK -> data_snapshots.data_snapshot_id
usage_purpose	VARCHAR(80)	NOT NULL; DEFAULT

risk_model_versions [CORE]

Versioned formula implementation for quantitative risk calculations.

PK: risk_model_version_id

FKs: created_by_user_id -> users.user_id

Column	Type	Declared flags / references
risk_model_version_id	UUID	PK; DEFAULT
model_key	VARCHAR(100)	NOT NULL
version_number	INTEGER	NOT NULL; CHECK
methodology	VARCHAR(40)	NOT NULL
implementation_version	VARCHAR(120)	NOT NULL
implementation_hash	CHAR(64)	-
formula_definitions	JSONB	NOT NULL; DEFAULT
created_by_user_id	UUID	FK -> users.user_id
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

UNIQUE (model_key, version_number)

risk_config_versions [CORE]

Versioned risk-free/benchmark/annualization/confidence defaults.

PK: risk_config_version_id

FKs: workspace_id -> workspaces.workspace_id; default_benchmark_instrument_id -> instruments.instrument_id; created_by_user_id -> users.user_id

Column	Type	Declared flags / references
risk_config_version_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id
config_name	VARCHAR(160)	NOT NULL
version_number	INTEGER	NOT NULL; CHECK
risk_free_rate	NUMERIC(14,10)	-
annualization_factor	NUMERIC(12,6)	NOT NULL; DEFAULT; CHECK
default_confidence_level	NUMERIC(6,5)	NOT NULL; DEFAULT; CHECK
default_benchmark_instrument_id	UUID	FK -> instruments.instrument_id
configuration	JSONB	NOT NULL; DEFAULT
is_active	BOOLEAN	NOT NULL; DEFAULT
created_by_user_id	UUID	FK -> users.user_id
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

UNIQUE (workspace_id, config_name, version_number)

risk_runs [CORE]

Reproducible historical/parametric/Monte Carlo run config, seed, data and limitations.

PK: risk_run_id

FKs: execution_run_id -> execution_runs.execution_run_id; workspace_id -> workspaces.workspace_id; owner_user_id -> users.user_id; portfolio_id -> portfolios.portfolio_id; portfolio_snapshot_id -> portfolio_snapshots.portfolio_snapshot_id; risk_model_version_id -> risk_model_versions.risk_model_version_id; risk_config_version_id -> risk_config_versions.risk_config_version_id; data_snapshot_id -> data_snapshots.data_snapshot_id; benchmark_instrument_id -> instruments.instrument_id

Column	Type	Declared flags / references
risk_run_id	UUID	PK; DEFAULT
execution_run_id	UUID	FK -> execution_runs.execution_run_id; NOT NULL; UNIQUE
workspace_id	UUID	FK -> workspaces.workspace_id; NOT NULL
owner_user_id	UUID	FK -> users.user_id; NOT NULL
portfolio_id	UUID	FK -> portfolios.portfolio_id
portfolio_snapshot_id	UUID	FK -> portfolio_snapshots.portfolio_snapshot_id
risk_model_version_id	UUID	FK -> risk_model_versions.risk_model_version_id; NOT NULL
risk_config_version_id	UUID	FK -> risk_config_versions.risk_config_version_id
data_snapshot_id	UUID	FK -> data_snapshots.data_snapshot_id; NOT NULL
benchmark_instrument_id	UUID	FK -> instruments.instrument_id
run_name	VARCHAR(180)	-
calculation_method	VARCHAR(24)	NOT NULL; CHECK
start_date	DATE	NOT NULL
end_date	DATE	NOT NULL
return_frequency	VARCHAR(20)	NOT NULL; DEFAULT
confidence_level	NUMERIC(6,5)	NOT NULL; DEFAULT; CHECK
annualization_factor	NUMERIC(12,6)	NOT NULL; DEFAULT; CHECK
risk_free_rate	NUMERIC(14,10)	-
simulation_count	INTEGER	-
random_seed	BIGINT	-
covariance_method	VARCHAR(40)	-
config_snapshot	JSONB	NOT NULL; DEFAULT
status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
data_limitations	JSONB	NOT NULL; DEFAULT
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
CHECK (end_date >= start_date)
CHECK (simulation_count IS NULL OR simulation_count > 0)
```

risk_run_positions [CORE]

Frozen portfolio weights/positions used by a risk calculation.

PK: risk_run_id, instrument_id

FKs: risk_run_id -> risk_runs.risk_run_id; instrument_id -> instruments.instrument_id; currency_code -> currencies.currency_code; source_portfolio_position_id -> portfolio_positions.portfolio_position_id

Column	Type	Declared flags / references
risk_run_id	UUID	PK; FK -> risk_runs.risk_run_id
instrument_id	UUID	PK; FK -> instruments.instrument_id
quantity	NUMERIC(28,10)	-
market_value	NUMERIC(30,10)	-
portfolio_weight	NUMERIC(18,10)	NOT NULL
currency_code	CHAR(3)	FK -> currencies.currency_code
source_portfolio_position_id	UUID	FK -> portfolio_positions.portfolio_position_id

risk_metric_results [CORE]

VaR/CVaR/volatility/beta/drawdown/concentration and other metrics.

PK: `risk_run_id, metric_key`

FKs: `risk_run_id -> risk_runs.risk_run_id`

Column	Type	Declared flags / references
risk_run_id	UUID	PK; FK -> risk_runs.risk_run_id
metric_key	VARCHAR(100)	PK
metric_value	NUMERIC(30,12)	-
unit	VARCHAR(40)	NOT NULL
method	VARCHAR(80)	NOT NULL
confidence_level	NUMERIC(6,5)	-
annualization_factor	NUMERIC(12,6)	-
assumptions	JSONB	NOT NULL; DEFAULT
limitation	TEXT	-

risk_covariance_pairs [CORE]

Covariance/correlation entries for instrument pairs.

PK: `risk_run_id, instrument_a_id, instrument_b_id`

FKs: `risk_run_id -> risk_runs.risk_run_id; instrument_a_id -> instruments.instrument_id; instrument_b_id -> instruments.instrument_id`

Column	Type	Declared flags / references
risk_run_id	UUID	PK; FK -> risk_runs.risk_run_id
instrument_a_id	UUID	PK; FK -> instruments.instrument_id
instrument_b_id	UUID	PK; FK -> instruments.instrument_id
covariance	NUMERIC(30,16)	-
correlation	NUMERIC(18,12)	-
observation_count	INTEGER	CHECK

Table-level constraints:

```
CHECK (instrument_a_id <= instrument_b_id)
CHECK (correlation IS NULL OR correlation BETWEEN -1 AND 1)
```

risk_time_series [CORE]

Daily returns, portfolio values, high-water marks and drawdowns.

PK: `risk_run_id, observation_date`

FKs: `risk_run_id -> risk_runs.risk_run_id; source_snapshot_id -> data_snapshots.data_snapshot_id`

Column	Type	Declared flags / references
risk_run_id	UUID	PK; FK -> risk_runs.risk_run_id
observation_date	DATE	PK
portfolio_return	NUMERIC(18,10)	-
benchmark_return	NUMERIC(18,10)	-
portfolio_value	NUMERIC(30,10)	-
drawdown	NUMERIC(18,10)	-
high_water_mark	NUMERIC(30,10)	-
rolling_volatility	NUMERIC(18,10)	-
source_snapshot_id	UUID	FK -> data_snapshots.data_snapshot_id

stress_scenarios [CORE]

Named historical/factor/custom shock definitions.

PK: stress_scenario_id

FKs: workspace_id -> workspaces.workspace_id; owner_user_id -> users.user_id

Column	Type	Declared flags / references
stress_scenario_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id
owner_user_id	UUID	FK -> users.user_id
scenario_name	VARCHAR(180)	NOT NULL
scenario_description	TEXT	-
scenario_type	VARCHAR(30)	NOT NULL; CHECK
shocks	JSONB	NOT NULL
is_system_scenario	BOOLEAN	NOT NULL; DEFAULT
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

UNIQUE (workspace_id, scenario_name)

risk_stress_results [CORE]

Run-by-scenario portfolio impact and limitations.

PK: risk_run_id, stress_scenario_id

FKs: risk_run_id -> risk_runs.risk_run_id; stress_scenario_id -> stress_scenarios.stress_scenario_id

Column	Type	Declared flags / references
risk_run_id	UUID	PK; FK -> risk_runs.risk_run_id
stress_scenario_id	UUID	PK; FK -> stress_scenarios.stress_scenario_id
pnl_amount	NUMERIC(30,10)	-
pnl_percent	NUMERIC(18,10)	-
stressed_portfolio_value	NUMERIC(30,10)	-
result_detail	JSONB	NOT NULL; DEFAULT
scenario_definition_snapshot	JSONB	NOT NULL; DEFAULT
limitation	TEXT	-

risk_distribution_points [CORE]

Quantile/distribution outputs for stochastic risk metrics.

PK: risk_run_id, metric_key, percentile

FKs: risk_run_id -> risk_runs.risk_run_id

Column	Type	Declared flags / references
risk_run_id	UUID	PK; FK -> risk_runs.risk_run_id
metric_key	VARCHAR(100)	PK
percentile	NUMERIC(8,6)	PK; CHECK
metric_value	NUMERIC(30,12)	NOT NULL

risk_simulation_points [CORE]

Stored Monte Carlo sample paths/steps, where retention is enabled.

PK: risk_run_id, path_number, step_number

FKs: risk_run_id -> risk_runs.risk_run_id

Column	Type	Declared flags / references
risk_run_id	UUID	PK; FK -> risk_runs.risk_run_id
path_number	INTEGER	PK; CHECK
step_number	INTEGER	PK; CHECK
observation_date	DATE	-
simulated_return	NUMERIC(18,10)	-
simulated_portfolio_value	NUMERIC(30,10)	-

reports [CORE]

Saved report metadata/content or artifact pointer, source snapshots and as-of time.

PK: report_id

FKs: workspace_id -> workspaces.workspace_id; owner_user_id -> users.user_id; execution_run_id -> execution_runs.execution_run_id

Column	Type	Declared flags / references
report_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id; NOT NULL
owner_user_id	UUID	FK -> users.user_id; NOT NULL
execution_run_id	UUID	FK -> execution_runs.execution_run_id
report_type	VARCHAR(40)	NOT NULL; CHECK
title	VARCHAR(300)	NOT NULL
summary	TEXT	-
report_content	JSONB	NOT NULL; DEFAULT
artifact_uri	TEXT	-
artifact_mime_type	VARCHAR(120)	-
as_of_at	TIMESTAMPTZ	-
report_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT
updated_at	TIMESTAMPTZ	NOT NULL; DEFAULT

report_citations [CORE]

Source-backed report citations to chunks, facts, versions and source URLs.

PK: report_citation_id

FKs: report_id -> reports.report_id; document_version_id -> document_versions.document_version_id; document_chunk_id -> document_chunks.document_chunk_id; financial_fact_id -> financial_facts.financial_fact_id; source_registry_id -> source_registry.source_registry_id

Column	Type	Declared flags / references
report_citation_id	UUID	PK; DEFAULT
report_id	UUID	FK -> reports.report_id; NOT NULL
document_version_id	UUID	FK -> document_versions.document_version_id
document_chunk_id	UUID	FK -> document_chunks.document_chunk_id
financial_fact_id	UUID	FK -> financial_facts.financial_fact_id
source_registry_id	UUID	FK -> source_registry.source_registry_id
citation_label	VARCHAR(300)	-
source_url	TEXT	-
page_number	INTEGER	-
section_label	TEXT	-
quoted_text	TEXT	-
accessed_at	TIMESTAMPTZ	-

Table-level constraints:

```
CHECK (num_nonnulls(document_version_id, document_chunk_id, financial_fact_id, source_registry_id, source_url) >= 1)
```

3.8 Screening, strategies, workflows and backtesting (26 tables)

Implementation scope
CORE TARGET SCHEMA - implement in staged releases according to dependencies and priorities.

saved_screens [CORE]

Saved financial screening rule trees.

PK: `saved_screen_id`

FKs: `workspace_id -> workspaces.workspace_id; owner_user_id -> users.user_id`

Column	Type	Declared flags / references
saved_screen_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id; NOT NULL
owner_user_id	UUID	FK -> users.user_id; NOT NULL
screen_name	VARCHAR(180)	NOT NULL
description	TEXT	-
market_code	VARCHAR(20)	NOT NULL; DEFAULT
rule_definition	JSONB	NOT NULL
version_number	INTEGER	NOT NULL; DEFAULT; CHECK
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT
updated_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (workspace_id, owner_user_id, screen_name)
```

screen_runs [CORE]

Version/data snapshot used for each screen execution.

PK: `screen_run_id`

FKs: `saved_screen_id -> saved_screens.saved_screen_id; execution_run_id -> execution_runs.execution_run_id; data_snapshot_id -> data_snapshots.data_snapshot_id`

Column	Type	Declared flags / references
screen_run_id	UUID	PK; DEFAULT
saved_screen_id	UUID	FK -> saved_screens.saved_screen_id; NOT NULL
execution_run_id	UUID	FK -> execution_runs.execution_run_id; NOT NULL; UNIQUE
data_snapshot_id	UUID	FK -> data_snapshots.data_snapshot_id; NOT NULL
run_parameters	JSONB	NOT NULL; DEFAULT
result_count	INTEGER	NOT NULL; DEFAULT; CHECK
status	VARCHAR(20)	NOT NULL; CHECK
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

screen_results [CORE]

Instruments that matched a screen and relevant metric values.

PK: `screen_run_id, instrument_id`

FKs: `screen_run_id -> screen_runs.screen_run_id; instrument_id -> instruments.instrument_id`

Column	Type	Declared flags / references
screen_run_id	UUID	PK; FK -> screen_runs.screen_run_id
instrument_id	UUID	PK; FK -> instruments.instrument_id
result_rank	INTEGER	-
matched_conditions	JSONB	NOT NULL; DEFAULT
metrics	JSONB	NOT NULL; DEFAULT
observed_at	TIMESTAMPTZ	NOT NULL; DEFAULT

strategies [CORE]

Strategy identity/ownership and lifecycle.

PK: `strategy_id`

FKs: `workspace_id -> workspaces.workspace_id; owner_user_id -> users.user_id`

Column	Type	Declared flags / references
strategy_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id; NOT NULL
owner_user_id	UUID	FK -> users.user_id; NOT NULL
strategy_name	VARCHAR(180)	NOT NULL
description	TEXT	-
strategy_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT
updated_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (workspace_id, owner_user_id, strategy_name)
```

strategy_versions [CORE]

Immutable entry/exit/sizing/risk rules with validation status.

PK: `strategy_version_id`

FKs: `strategy_id -> strategies.strategy_id; parent_version_id -> strategy_versions.strategy_version_id; created_by_user_id -> users.user_id`

Column	Type	Declared flags / references
strategy_version_id	UUID	PK; DEFAULT
strategy_id	UUID	FK -> strategies.strategy_id; NOT NULL
version_number	INTEGER	NOT NULL; CHECK
parent_version_id	UUID	FK -> strategy_versions.strategy_version_id
strategy_definition	JSONB	NOT NULL
entry_rules	JSONB	NOT NULL
exit_rules	JSONB	NOT NULL
position_sizing_rules	JSONB	NOT NULL; DEFAULT
risk_controls	JSONB	NOT NULL; DEFAULT
parameter_schema	JSONB	NOT NULL; DEFAULT
validation_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
validation_report	JSONB	NOT NULL; DEFAULT
created_by_user_id	UUID	FK -> users.user_id
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (strategy_id, version_number)
```

strategy_universe_instruments [CORE]

Explicit instruments in a strategy universe.

PK: `strategy_version_id, instrument_id`

FKs: `strategy_version_id -> strategy_versions.strategy_version_id; instrument_id -> instruments.instrument_id`

Column	Type	Declared flags / references
strategy_version_id	UUID	PK; FK -> strategy_versions.strategy_version_id
instrument_id	UUID	PK; FK -> instruments.instrument_id

strategy_variants [CORE]

Named parameter variants for repeatable comparison.

PK: strategy_variant_id

FKs: strategy_version_id -> strategy_versions.strategy_version_id; created_by_user_id -> users.user_id

Column	Type	Declared flags / references
strategy_variant_id	UUID	PK; DEFAULT
strategy_version_id	UUID	FK -> strategy_versions.strategy_version_id; NOT NULL
variant_name	VARCHAR(160)	NOT NULL
parameter_overrides	JSONB	NOT NULL
created_by_user_id	UUID	FK -> users.user_id
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

UNIQUE (strategy_version_id, variant_name)

signals [CORE]

Generated signal with rule evidence, data snapshot and status.

PK: signal_id

FKs: workspace_id -> workspaces.workspace_id; instrument_id -> instruments.instrument_id; strategy_version_id -> strategy_versions.strategy_version_id; screen_run_id -> screen_runs.screen_run_id; data_snapshot_id -> data_snapshots.data_snapshot_id

Column	Type	Declared flags / references
signal_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id; NOT NULL
instrument_id	UUID	FK -> instruments.instrument_id; NOT NULL
strategy_version_id	UUID	FK -> strategy_versions.strategy_version_id
screen_run_id	UUID	FK -> screen_runs.screen_run_id
data_snapshot_id	UUID	FK -> data_snapshots.data_snapshot_id; NOT NULL
signal_type	VARCHAR(40)	NOT NULL
direction	VARCHAR(20)	CHECK
explanation	TEXT	-
triggering_rule	JSONB	NOT NULL
observed_inputs	JSONB	NOT NULL; DEFAULT
generated_at	TIMESTAMPTZ	NOT NULL; DEFAULT
current_status	VARCHAR(24)	NOT NULL; DEFAULT; CHECK

signal_observations [CORE]

Metric-level observations backing a signal.

PK: signal_observation_id

FKs: signal_id -> signals.signal_id; source_price_bar_id -> price_bars.price_bar_id; source_financial_fact_id -> financial_facts.financial_fact_id

Column	Type	Declared flags / references
signal_observation_id	UUID	PK; DEFAULT
signal_id	UUID	FK -> signals.signal_id; NOT NULL
metric_key	VARCHAR(100)	NOT NULL
numeric_value	NUMERIC(30,12)	-
text_value	TEXT	-
unit	VARCHAR(40)	-
source_price_bar_id	UUID	FK -> price_bars.price_bar_id
source_financial_fact_id	UUID	FK -> financial_facts.financial_fact_id
observed_at	TIMESTAMPTZ	NOT NULL; DEFAULT
observation_detail	JSONB	NOT NULL; DEFAULT

signal_status_history [CORE]

Signal status-change history.

PK: signal_status_history_id

FKs: signal_id -> signals.signal_id; changed_by_user_id -> users.user_id

Column	Type	Declared flags / references
signal_status_history_id	UUID	PK; DEFAULT
signal_id	UUID	FK -> signals.signal_id; NOT NULL
previous_status	VARCHAR(24)	-
new_status	VARCHAR(24)	NOT NULL
changed_by_user_id	UUID	FK -> users.user_id
reason	TEXT	-
changed_at	TIMESTAMPTZ	NOT NULL; DEFAULT

workflow_definitions [CORE]

Owned workflow identity and lifecycle.

PK: workflow_definition_id

FKs: workspace_id -> workspaces.workspace_id; owner_user_id -> users.user_id

Column	Type	Declared flags / references
workflow_definition_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id; NOT NULL
owner_user_id	UUID	FK -> users.user_id; NOT NULL
workflow_name	VARCHAR(180)	NOT NULL
description	TEXT	-
workflow_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT
updated_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (workspace_id, owner_user_id, workflow_name)
```

workflow_versions [CORE]

Versioned instructions, model config, schema, retry and policy.

PK: workflow_version_id

FKs: workflow_definition_id -> workflow_definitions.workflow_definition_id; ai_model_id -> ai_models.ai_model_id; ai_config_version_id -> ai_config_versions.ai_config_version_id; created_by_user_id -> users.user_id

Column	Type	Declared flags / references
workflow_version_id	UUID	PK; DEFAULT
workflow_definition_id	UUID	FK -> workflow_definitions.workflow_definition_id; NOT NULL
version_number	INTEGER	NOT NULL; CHECK
instructions	TEXT	-
ai_model_id	UUID	FK -> ai_models.ai_model_id
ai_config_version_id	UUID	FK -> ai_config_versions.ai_config_version_id
input_schema	JSONB	NOT NULL; DEFAULT
execution_policy	JSONB	NOT NULL; DEFAULT
retry_policy	JSONB	NOT NULL; DEFAULT
timeout_seconds	INTEGER	NOT NULL; DEFAULT; CHECK
is_published	BOOLEAN	NOT NULL; DEFAULT
created_by_user_id	UUID	FK -> users.user_id
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (workflow_definition_id, version_number)
```

workflow_steps [CORE]

Ordered workflow nodes and step configuration.

PK: `workflow_step_id`

FKs: `workflow_version_id -> workflow_versions.workflow_version_id`

Column	Type	Declared flags / references
workflow_step_id	UUID	PK; DEFAULT
workflow_version_id	UUID	FK -> workflow_versions.workflow_version_id; NOT NULL
step_key	VARCHAR(100)	NOT NULL
step_order	INTEGER	NOT NULL; CHECK
step_type	VARCHAR(40)	NOT NULL; CHECK
step_config	JSONB	NOT NULL
timeout_seconds	INTEGER	CHECK
retry_limit	SMALLINT	NOT NULL; DEFAULT; CHECK

Table-level constraints:

```
UNIQUE (workflow_version_id, step_key)
UNIQUE (workflow_version_id, step_order)
```

workflow_tool_bindings [CORE]

Allowlisted tools attached to a workflow version.

PK: `workflow_version_id, tool_registry_id`

FKs: `workflow_version_id -> workflow_versions.workflow_version_id; tool_registry_id -> tool_registry.tool_registry_id`

Column	Type	Declared flags / references
workflow_version_id	UUID	PK; FK -> workflow_versions.workflow_version_id
tool_registry_id	UUID	PK; FK -> tool_registry.tool_registry_id
allowed_operations	TEXT[]	NOT NULL; DEFAULT

workflow_collection_bindings [CORE]

Authorized research collections available to a workflow.

PK: `workflow_version_id, research_collection_id`

FKs: `workflow_version_id -> workflow_versions.workflow_version_id; research_collection_id -> research_collections.research_collection_id`

Column	Type	Declared flags / references
workflow_version_id	UUID	PK; FK -> workflow_versions.workflow_version_id
research_collection_id	UUID	PK; FK -> research_collections.research_collection_id

workflow_data_source_bindings [CORE]

Data source and allowed-operation bindings for workflows.

PK: `workflow_version_id, structured_data_source_id`

FKs: `workflow_version_id -> workflow_versions.workflow_version_id; structured_data_source_id -> structured_data_sources.structured_data_source_id`

Column	Type	Declared flags / references
workflow_version_id	UUID	PK; FK -> workflow_versions.workflow_version_id
structured_data_source_id	UUID	PK; FK -> structured_data_sources.structured_data_source_id
allowed_objects	JSONB	NOT NULL; DEFAULT
allowed_operations	TEXT[]	NOT NULL; DEFAULT

workflow_schedules [CORE]

Timezone-aware workflow schedules.

PK: workflow_schedule_id

FKs: workflow_definition_id -> workflow_definitions.workflow_definition_id; workflow_version_id -> workflow_versions.workflow_version_id; created_by_user_id -> users.user_id

Column	Type	Declared flags / references
workflow_schedule_id	UUID	PK; DEFAULT
workflow_definition_id	UUID	FK -> workflow_definitions.workflow_definition_id; NOT NULL
workflow_version_id	UUID	FK -> workflow_versions.workflow_version_id; NOT NULL
schedule_expression	VARCHAR(120)	NOT NULL
timezone_name	VARCHAR(80)	NOT NULL
schedule_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
next_run_at	TIMESTAMPTZ	-
last_run_at	TIMESTAMPTZ	-
created_by_user_id	UUID	FK -> users.user_id
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

workflow_runs [CORE]

Workflow-version to execution-run link and run I/O.

PK: workflow_run_id

FKs: workflow_definition_id -> workflow_definitions.workflow_definition_id; workflow_version_id -> workflow_versions.workflow_version_id; execution_run_id -> execution_runs.execution_run_id; workflow_schedule_id -> workflow_schedules.workflow_schedule_id

Column	Type	Declared flags / references
workflow_run_id	UUID	PK; DEFAULT
workflow_definition_id	UUID	FK -> workflow_definitions.workflow_definition_id; NOT NULL
workflow_version_id	UUID	FK -> workflow_versions.workflow_version_id; NOT NULL
execution_run_id	UUID	FK -> execution_runs.execution_run_id; NOT NULL; UNIQUE
workflow_schedule_id	UUID	FK -> workflow_schedules.workflow_schedule_id
input_payload	JSONB	NOT NULL; DEFAULT
output_summary	JSONB	NOT NULL; DEFAULT
validation_result	JSONB	NOT NULL; DEFAULT
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

workflow_approvals [CORE]

Human approval requests and decisions.

PK: workflow_approval_id

FKs: workflow_run_id -> workflow_runs.workflow_run_id; workflow_step_id -> workflow_steps.workflow_step_id; requested_from_user_id -> users.user_id; decision_by_user_id -> users.user_id

Column	Type	Declared flags / references
workflow_approval_id	UUID	PK; DEFAULT
workflow_run_id	UUID	FK -> workflow_runs.workflow_run_id; NOT NULL
workflow_step_id	UUID	FK -> workflow_steps.workflow_step_id
requested_from_user_id	UUID	FK -> users.user_id
decision_by_user_id	UUID	FK -> users.user_id
decision	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
requested_action	JSONB	NOT NULL
decision_note	TEXT	-
requested_at	TIMESTAMPTZ	NOT NULL; DEFAULT
decided_at	TIMESTAMPTZ	-

cost_model_versions [CORE]

Versioned commission/slippage/tax/transaction cost assumptions.

PK: `cost_model_version_id`

FKs: `workspace_id -> workspaces.workspace_id; created_by_user_id -> users.user_id`

Column	Type	Declared flags / references
cost_model_version_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id
model_name	VARCHAR(160)	NOT NULL
version_number	INTEGER	NOT NULL; CHECK
market_code	VARCHAR(20)	NOT NULL; DEFAULT
assumptions	JSONB	NOT NULL
created_by_user_id	UUID	FK -> users.user_id
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (workspace_id, model_name, version_number)
```

backtest_runs [CORE]

Strategy-version/backtest config and source snapshot for reproducibility.

PK: `backtest_run_id`

FKs: `execution_run_id -> execution_runs.execution_run_id; workspace_id -> workspaces.workspace_id; owner_user_id -> users.user_id; strategy_version_id -> strategy_versions.strategy_version_id; strategy_variant_id -> strategy_variants.strategy_variant_id; data_snapshot_id -> data_snapshots.data_snapshot_id; cost_model_version_id -> cost_model_versions.cost_model_version_id; benchmark_instrument_id -> instruments.instrument_id; currency_code -> currencies.currency_code`

Column	Type	Declared flags / references
backtest_run_id	UUID	PK; DEFAULT
execution_run_id	UUID	FK -> execution_runs.execution_run_id; NOT NULL; UNIQUE
workspace_id	UUID	FK -> workspaces.workspace_id; NOT NULL
owner_user_id	UUID	FK -> users.user_id; NOT NULL
strategy_version_id	UUID	FK -> strategy_versions.strategy_version_id; NOT NULL
strategy_variant_id	UUID	FK -> strategy_variants.strategy_variant_id
data_snapshot_id	UUID	FK -> data_snapshots.data_snapshot_id; NOT NULL
cost_model_version_id	UUID	FK -> cost_model_versions.cost_model_version_id
benchmark_instrument_id	UUID	FK -> instruments.instrument_id
start_date	DATE	NOT NULL
end_date	DATE	NOT NULL
interval_code	VARCHAR(20)	NOT NULL; DEFAULT
initial_capital	NUMERIC(30,10)	NOT NULL; CHECK
currency_code	CHAR(3)	FK -> currencies.currency_code; NOT NULL
execution_assumptions	JSONB	NOT NULL; DEFAULT
integrity_checks	JSONB	NOT NULL; DEFAULT
out_of_sample_config	JSONB	NOT NULL; DEFAULT
status	VARCHAR(24)	NOT NULL; DEFAULT; CHECK
data_limitations	JSONB	NOT NULL; DEFAULT
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
CHECK (end_date >= start_date)
```

backtest_metrics [CORE]

Named metric outputs from a specific backtest.

PK: `backtest_run_id, metric_key`

FKs: `backtest_run_id -> backtest_runs.backtest_run_id`

Column	Type	Declared flags / references
backtest_run_id	UUID	PK; FK -> backtest_runs.backtest_run_id
metric_key	VARCHAR(100)	PK
metric_value	NUMERIC(30,12)	-
unit	VARCHAR(40)	NOT NULL
definition_version	VARCHAR(100)	-
metric_detail	JSONB	NOT NULL; DEFAULT

backtest_equity_points [CORE]

Equity/benchmark curve, returns, drawdown and exposure series.

PK: backtest_run_id, observation_date

FKs: backtest_run_id -> backtest_runs.backtest_run_id

Column	Type	Declared flags / references
backtest_run_id	UUID	PK; FK -> backtest_runs.backtest_run_id
observation_date	DATE	PK
portfolio_value	NUMERIC(30,10)	-
benchmark_value	NUMERIC(30,10)	-
period_return	NUMERIC(18,10)	-
drawdown	NUMERIC(18,10)	-
exposure	NUMERIC(18,10)	-

backtest_trades [CORE]

Trade-level entry/exit/cost/P&L/trigger records.

PK: backtest_trade_id

FKs: backtest_run_id -> backtest_runs.backtest_run_id; instrument_id -> instruments.instrument_id; entry_signal_id -> signals.signal_id; exit_signal_id -> signals.signal_id

Column	Type	Declared flags / references
backtest_trade_id	UUID	PK; DEFAULT
backtest_run_id	UUID	FK -> backtest_runs.backtest_run_id; NOT NULL
instrument_id	UUID	FK -> instruments.instrument_id; NOT NULL
entry_signal_id	UUID	FK -> signals.signal_id
exit_signal_id	UUID	FK -> signals.signal_id
direction	VARCHAR(12)	NOT NULL; CHECK
entry_at	TIMESTAMPTZ	NOT NULL
exit_at	TIMESTAMPTZ	-
quantity	NUMERIC(28,10)	NOT NULL
entry_price	NUMERIC(24,10)	NOT NULL
exit_price	NUMERIC(24,10)	-
gross_pnl	NUMERIC(30,10)	-
fees	NUMERIC(24,10)	NOT NULL; DEFAULT
slippage_cost	NUMERIC(24,10)	NOT NULL; DEFAULT
net_pnl	NUMERIC(30,10)	-
holding_period_seconds	BIGINT	-
trade_detail	JSONB	NOT NULL; DEFAULT

Table-level constraints:

```
CHECK (exit_at IS NULL OR exit_at >= entry_at)
```

backtest_run_comparisons [CORE]

Saved comparison group for multiple backtests.

PK: backtest_run_comparison_id

FKs: workspace_id -> workspaces.workspace_id; owner_user_id -> users.user_id

Column	Type	Declared flags / references
backtest_run_comparison_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id; NOT NULL
owner_user_id	UUID	FK -> users.user_id; NOT NULL
comparison_name	VARCHAR(180)	NOT NULL
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (workspace_id, owner_user_id, comparison_name)
```

backtest_comparison_members [CORE]

Backtest runs in comparison groups.

PK: backtest_run_comparison_id, backtest_run_id

FKs: backtest_run_comparison_id -> backtest_run_comparisons.backtest_run_comparison_id; backtest_run_id -> backtest_runs.backtest_run_id

Column	Type	Declared flags / references
backtest_run_comparison_id	UUID	PK; FK -> backtest_run_comparisons.backtest_run_comparison_id
backtest_run_id	UUID	PK; FK -> backtest_runs.backtest_run_id
display_order	INTEGER	NOT NULL; DEFAULT

3.9 Optional paper trading and broker integrations (15 tables)

> **Implementation scope**
> OPTIONAL PHASE 2+ - do not apply in Release 1.

broker_providers [OPTIONAL]

Optional broker adapters and declared account/order modes.

PK: broker_provider_id

FKs: source_registry_id -> source_registry.source_registry_id

Column	Type	Declared flags / references
broker_provider_id	UUID	PK; DEFAULT
source_registry_id	UUID	FK -> source_registry.source_registry_id
provider_key	VARCHAR(100)	NOT NULL; UNIQUE
display_name	VARCHAR(160)	NOT NULL
supported_markets	TEXT[]	NOT NULL; DEFAULT
supports_read_only	BOOLEAN	NOT NULL; DEFAULT
supports_paper	BOOLEAN	NOT NULL; DEFAULT
supports_live_orders	BOOLEAN	NOT NULL; DEFAULT
adapter_name	VARCHAR(120)	NOT NULL
enabled	BOOLEAN	NOT NULL; DEFAULT
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

broker_connections [OPTIONAL]

Optional user-owned broker connection; credentials stored externally.

PK: broker_connection_id

FKs: broker_provider_id -> broker_providers.broker_provider_id; workspace_id -> workspaces.workspace_id; owner_user_id -> users.user_id

Column	Type	Declared flags / references
broker_connection_id	UUID	PK; DEFAULT
broker_provider_id	UUID	FK -> broker_providers.broker_provider_id; NOT NULL
workspace_id	UUID	FK -> workspaces.workspace_id; NOT NULL
owner_user_id	UUID	FK -> users.user_id; NOT NULL
connection_name	VARCHAR(160)	NOT NULL
secret_ref	TEXT	NOT NULL
permission_mode	VARCHAR(24)	NOT NULL; DEFAULT; CHECK
connection_status	VARCHAR(24)	NOT NULL; DEFAULT; CHECK
last_tested_at	TIMESTAMPTZ	-
last_successful_sync_at	TIMESTAMPTZ	-
non_secret_config	JSONB	NOT NULL; DEFAULT
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

UNIQUE (workspace_id, owner_user_id, connection_name)

broker_accounts [OPTIONAL]

Accounts exposed by a broker connection.

PK: broker_account_id

FKs: broker_connection_id -> broker_connections.broker_connection_id; currency_code -> currencies.currency_code

Column	Type	Declared flags / references
broker_account_id	UUID	PK; DEFAULT
broker_connection_id	UUID	FK -> broker_connections.broker_connection_id; NOT NULL
external_account_id	VARCHAR(240)	NOT NULL
account_name	VARCHAR(160)	-
account_type	VARCHAR(40)	-
account_mode	VARCHAR(20)	NOT NULL; CHECK
currency_code	CHAR(3)	FK -> currencies.currency_code; NOT NULL
account_status	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
last_reconciled_at	TIMESTAMPTZ	-
account_metadata	JSONB	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (broker_connection_id, external_account_id)
```

paper_trading_accounts [OPTIONAL]

Paper-simulation setup linked to a portfolio.

PK: paper_trading_account_id

FKs: portfolio_id -> portfolios.portfolio_id; currency_code -> currencies.currency_code

Column	Type	Declared flags / references
paper_trading_account_id	UUID	PK; DEFAULT
portfolio_id	UUID	FK -> portfolios.portfolio_id; NOT NULL; UNIQUE
initial_cash	NUMERIC(30,10)	NOT NULL; CHECK
currency_code	CHAR(3)	FK -> currencies.currency_code; NOT NULL
simulation_config	JSONB	NOT NULL; DEFAULT
started_at	TIMESTAMPTZ	NOT NULL; DEFAULT
ended_at	TIMESTAMPTZ	-

Table-level constraints:

```
CHECK (ended_at IS NULL OR ended_at >= started_at)
```

broker_position_snapshots [OPTIONAL]

Point-in-time account snapshot from a broker.

PK: broker_position_snapshot_id

FKs: broker_account_id -> broker_accounts.broker_account_id; data_snapshot_id -> data_snapshots.data_snapshot_id; currency_code -> currencies.currency_code

Column	Type	Declared flags / references
broker_position_snapshot_id	UUID	PK; DEFAULT
broker_account_id	UUID	FK -> broker_accounts.broker_account_id; NOT NULL
data_snapshot_id	UUID	FK -> data_snapshots.data_snapshot_id
as_of_at	TIMESTAMPTZ	NOT NULL
total_equity	NUMERIC(30,10)	-
cash_balance	NUMERIC(30,10)	-
currency_code	CHAR(3)	FK -> currencies.currency_code; NOT NULL
provider_state	JSONB	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (broker_account_id, as_of_at)
```

broker_positions [OPTIONAL]

Holdings inside a broker position snapshot.

PK: broker_position_id

FKs: broker_position_snapshot_id -> broker_position_snapshots.broker_position_snapshot_id; instrument_id -> instruments.instrument_id; currency_code -> currencies.currency_code

Column	Type	Declared flags / references
broker_position_id	UUID	PK; DEFAULT
broker_position_snapshot_id	UUID	FK -> broker_position_snapshots.broker_position_snapshot_id; NOT NULL
instrument_id	UUID	FK -> instruments.instrument_id; NOT NULL
quantity	NUMERIC(28,10)	NOT NULL
average_cost	NUMERIC(24,10)	-
market_price	NUMERIC(24,10)	-
market_value	NUMERIC(30,10)	-
unrealized_pnl	NUMERIC(30,10)	-
currency_code	CHAR(3)	FK -> currencies.currency_code

Table-level constraints:

```
UNIQUE (broker_position_snapshot_id, instrument_id)
```

trading_safety_policies [OPTIONAL]

Fail-closed order permissions, limits and confirmation requirements.

PK: trading_safety_policy_id

FKs: workspace_id -> workspaces.workspace_id; updated_by_user_id -> users.user_id

Column	Type	Declared flags / references
trading_safety_policy_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id; NOT NULL; UNIQUE
allow_paper_orders	BOOLEAN	NOT NULL; DEFAULT
allow_live_orders	BOOLEAN	NOT NULL; DEFAULT
require_human_confirmation	BOOLEAN	NOT NULL; DEFAULT
require_step_up_auth	BOOLEAN	NOT NULL; DEFAULT
max_order_notional	NUMERIC(30,10)	-
max_position_weight	NUMERIC(18,10)	-
max_open_orders	INTEGER	-
policy_version	INTEGER	NOT NULL; DEFAULT; CHECK
updated_by_user_id	UUID	FK -> users.user_id
updated_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
CHECK (max_order_notional IS NULL OR max_order_notional > 0)
CHECK (max_position_weight IS NULL OR (max_position_weight > 0 AND max_position_weight <= 1))
CHECK (max_open_orders IS NULL OR max_open_orders > 0)
```

trading_orders [OPTIONAL]

Optional paper/live order lifecycle; live is disabled by default.

PK: `trading_order_id`

FKs: `portfolio_id -> portfolios.portfolio_id; broker_account_id -> broker_accounts.broker_account_id; workspace_id -> workspaces.workspace_id; submitted_by_user_id -> users.user_id; instrument_id -> instruments.instrument_id; strategy_version_id -> strategy_versions.strategy_version_id`

Column	Type	Declared flags / references
trading_order_id	UUID	PK; DEFAULT
portfolio_id	UUID	FK -> portfolios.portfolio_id
broker_account_id	UUID	FK -> broker_accounts.broker_account_id
workspace_id	UUID	FK -> workspaces.workspace_id; NOT NULL
submitted_by_user_id	UUID	FK -> users.user_id; NOT NULL
instrument_id	UUID	FK -> instruments.instrument_id; NOT NULL
strategy_version_id	UUID	FK -> strategy_versions.strategy_version_id
client_order_id	VARCHAR(240)	NOT NULL
provider_order_id	VARCHAR(240)	-
execution_mode	VARCHAR(12)	NOT NULL; CHECK
side	VARCHAR(8)	NOT NULL; CHECK
order_type	VARCHAR(20)	NOT NULL; CHECK
quantity	NUMERIC(28,10)	NOT NULL; CHECK
limit_price	NUMERIC(24,10)	-
stop_price	NUMERIC(24,10)	-
requested_notional	NUMERIC(30,10)	-
time_in_force	VARCHAR(20)	NOT NULL; DEFAULT
order_status	VARCHAR(24)	NOT NULL; DEFAULT; CHECK
confirmation_snapshot	JSONB	-
confirmed_at	TIMESTAMPTZ	-
safety_policy_snapshot	JSONB	NOT NULL; DEFAULT
idempotency_key	VARCHAR(240)	NOT NULL; UNIQUE
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT
updated_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
CHECK (order_type NOT IN ('limit','stop_limit') OR limit_price IS NOT NULL)
CHECK (order_type NOT IN ('stop','stop_limit') OR stop_price IS NOT NULL)
CHECK (execution_mode <> 'live' OR (broker_account_id IS NOT NULL AND confirmed_at IS NOT NULL))
```

trading_risk_checks [OPTIONAL]

Pre-trade risk-check results per order.

PK: `trading_risk_check_id`

FKs: `trading_order_id -> trading_orders.trading_order_id`

Column	Type	Declared flags / references
trading_risk_check_id	UUID	PK; DEFAULT
trading_order_id	UUID	FK -> trading_orders.trading_order_id; NOT NULL
passed	BOOLEAN	NOT NULL
blocking	BOOLEAN	NOT NULL; DEFAULT
input_snapshot	JSONB	NOT NULL
result_detail	JSONB	NOT NULL; DEFAULT

Table-level constraints:

```
check_key VARCHAR(100) NOT NULL
check_version VARCHAR(80) NOT NULL
checked_at TIMESTAMPTZ NOT NULL DEFAULT now()
UNIQUE (trading_order_id, check_key, check_version)
```

trading_order_approvals [OPTIONAL]

Explicit user decision for an order and displayed order snapshot.

PK: `trading_order_approval_id`

FKs: `trading_order_id -> trading_orders.trading_order_id; requested_from_user_id -> users.user_id; decided_by_user_id -> users.user_id`

Column	Type	Declared flags / references
trading_order_approval_id	UUID	PK; DEFAULT
trading_order_id	UUID	FK -> trading_orders.trading_order_id; NOT NULL
requested_from_user_id	UUID	FK -> users.user_id
decided_by_user_id	UUID	FK -> users.user_id
decision	VARCHAR(20)	NOT NULL; DEFAULT; CHECK
displayed_order_snapshot	JSONB	NOT NULL
decision_note	TEXT	-
requested_at	TIMESTAMPTZ	NOT NULL; DEFAULT
decided_at	TIMESTAMPTZ	-

trading_order_events [OPTIONAL]

Provider/local order state transition history.

PK: `trading_order_event_id`

FKs: `trading_order_id -> trading_orders.trading_order_id`

Column	Type	Declared flags / references
trading_order_event_id	UUID	PK; DEFAULT
trading_order_id	UUID	FK -> trading_orders.trading_order_id; NOT NULL
event_type	VARCHAR(60)	NOT NULL
previous_status	VARCHAR(24)	-
new_status	VARCHAR(24)	-
provider_event_id	VARCHAR(240)	-
event_detail_redacted	JSONB	NOT NULL; DEFAULT
occurred_at	TIMESTAMPTZ	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (trading_order_id, provider_event_id)
```

trading_order_fills [OPTIONAL]

Individual reported/simulated fill records.

PK: `trading_order_fill_id`

FKs: `trading_order_id -> trading_orders.trading_order_id; currency_code -> currencies.currency_code`

Column	Type	Declared flags / references
trading_order_fill_id	UUID	PK; DEFAULT
trading_order_id	UUID	FK -> trading_orders.trading_order_id; NOT NULL
provider_fill_id	VARCHAR(240)	-
fill_time	TIMESTAMPTZ	NOT NULL
quantity	NUMERIC(28,10)	NOT NULL; CHECK
fill_price	NUMERIC(24,10)	NOT NULL; CHECK
fee_amount	NUMERIC(24,10)	NOT NULL; DEFAULT; CHECK
currency_code	CHAR(3)	FK -> currencies.currency_code; NOT NULL
fill_metadata	JSONB	NOT NULL; DEFAULT

Table-level constraints:

```
UNIQUE (trading_order_id, provider_fill_id)
```

broker_reconciliations [OPTIONAL]

Local-vs-broker state reconciliation run.

PK: broker_reconciliation_id

FKs: broker_account_id -> broker_accounts.broker_account_id; local_snapshot_id -> broker_position_snapshots.broker_position_snapshot_id

Column	Type	Declared flags / references
broker_reconciliation_id	UUID	PK; DEFAULT
broker_account_id	UUID	FK -> broker_accounts.broker_account_id; NOT NULL
local_snapshot_id	UUID	FK -> broker_position_snapshots.broker_position_snapshot_id
status	VARCHAR(20)	NOT NULL; CHECK
started_at	TIMESTAMPTZ	NOT NULL; DEFAULT
finished_at	TIMESTAMPTZ	-
summary	JSONB	NOT NULL; DEFAULT
error_redacted	TEXT	-

broker_reconciliation_items [OPTIONAL]

Order/position/cash-level differences in reconciliation.

PK: broker_reconciliation_item_id

FKs: broker_reconciliation_id -> broker_reconciliations.broker_reconciliation_id; resolved_by_user_id -> users.user_id

Column	Type	Declared flags / references
broker_reconciliation_item_id	UUID	PK; DEFAULT
broker_reconciliation_id	UUID	FK -> broker_reconciliations.broker_reconciliation_id; NOT NULL
item_type	VARCHAR(40)	NOT NULL; CHECK
local_reference_id	UUID	-
provider_reference_id	VARCHAR(240)	-
reconciliation_status	VARCHAR(24)	NOT NULL; CHECK
local_state	JSONB	NOT NULL; DEFAULT
provider_state	JSONB	NOT NULL; DEFAULT
resolved_by_user_id	UUID	FK -> users.user_id
resolved_at	TIMESTAMPTZ	-

trade_journal_entries [OPTIONAL]

Trade rationale and post-trade notes.

PK: trade_journal_entry_id

FKs: workspace_id -> workspaces.workspace_id; owner_user_id -> users.user_id; trading_order_id -> trading_orders.trading_order_id; portfolio_transaction_id -> portfolio_transactions.portfolio_transaction_id; instrument_id -> instruments.instrument_id

Column	Type	Declared flags / references
trade_journal_entry_id	UUID	PK; DEFAULT
workspace_id	UUID	FK -> workspaces.workspace_id; NOT NULL
owner_user_id	UUID	FK -> users.user_id; NOT NULL
trading_order_id	UUID	FK -> trading_orders.trading_order_id
portfolio_transaction_id	UUID	FK -> portfolio_transactions.portfolio_transaction_id
instrument_id	UUID	FK -> instruments.instrument_id
rationale	TEXT	-
tags	TEXT[]	NOT NULL; DEFAULT
post_trade_observations	TEXT	-
created_at	TIMESTAMPTZ	NOT NULL; DEFAULT
updated_at	TIMESTAMPTZ	NOT NULL; DEFAULT

4. Relationship and implementation notes

These notes describe the intended relationship patterns recorded in the design package. They are not substitutes for service-layer authorization or financial calculation tests.

Data snapshots and provenance

Price bars, fundamentals, and saved analytical runs should preserve the applicable data_snapshot_id and source lineage. Do not silently replace historical run inputs with the latest available data.

DCF valuation

Valuation runs reference a valuation model version and retain assumptions and their provenance. The Python/backend calculation service is responsible for WACC, forecast cash flow, terminal value, enterprise/equity value and per-share value; the LLM explains the result.

Quantitative risk

Risk runs should preserve portfolio-position snapshots, the period and return frequency, risk model/configuration, confidence level, annualization convention, covariance method, simulation count and random seed where relevant. Missing inputs should yield an explicit limitation.

RAG and vector reconciliation

PostgreSQL owns document identity, permissions, versioning and lifecycle metadata. pgvector stores embeddings tied to stable chunk IDs. Re-indexing and deletion need idempotent background processing and reconciliation for partial failures.

Source licensing

Provider coverage and a successful local connection do not imply permission to store, display, export or redistribute all market data. Review source terms and entitlements before production.

Secrets

The SQL stores references/hashes instead of raw provider secrets. Keep credentials in a secrets manager and do not commit tokens to source control or seed data.

Optional trading

The optional trading schema is a separate extension. Real-money orders are outside Release 1 and should stay disabled until separate approval, security review, broker entitlement checks and fail-closed tests are completed.

Static validation summary

Table definitions parsed from SQL: 173 total (158 core and 15 optional). Catalogue entries matched all parsed SQL table names: True. Each declared FK target is documented in the generated table inventory. This validation does not execute DDL against a live PostgreSQL server and does not replace migration/integration testing.

Implementation files

`schema.sql` - core schema
`schema_optional_trading.sql` - optional trading extension
`seed_reference_data.sql` - development reference seed data
`table_catalog.md` - original human-readable PK/FK catalogue
`erd/ERD_*.svg` - vector ER diagrams
`erd/ERD_*.mmd` - editable Mermaid sources